

Expecting the unexpected Anomaly Detection

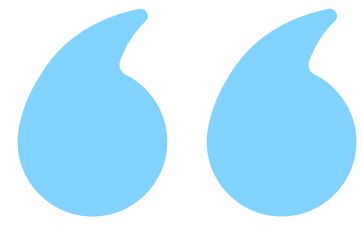
Dr. Ellen Rushe
Assistant Professor
School of Computing
Dublin City University

L2.24 (McNulty Building)
ellen.rushe@dcu.ie



**What is anomaly
detection?**





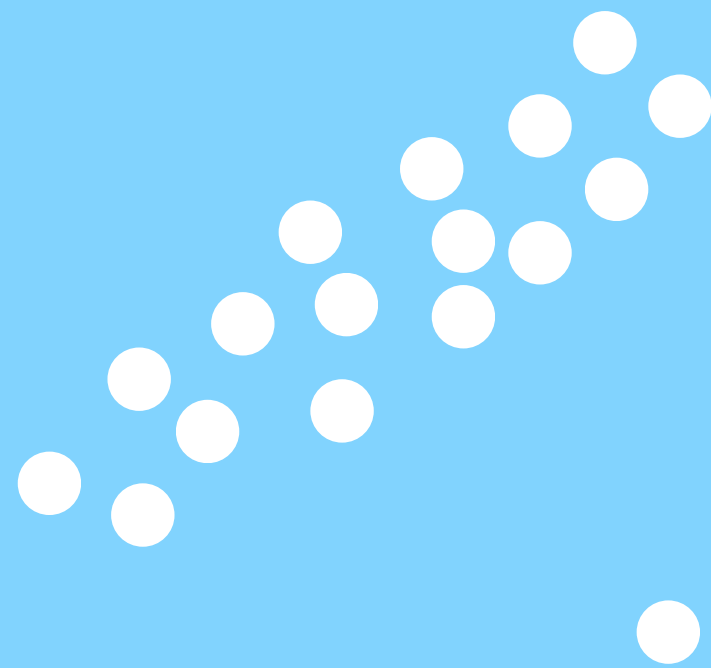
*Anomaly detection refers to the problem of finding patterns in data that **do not conform to expected behavior***



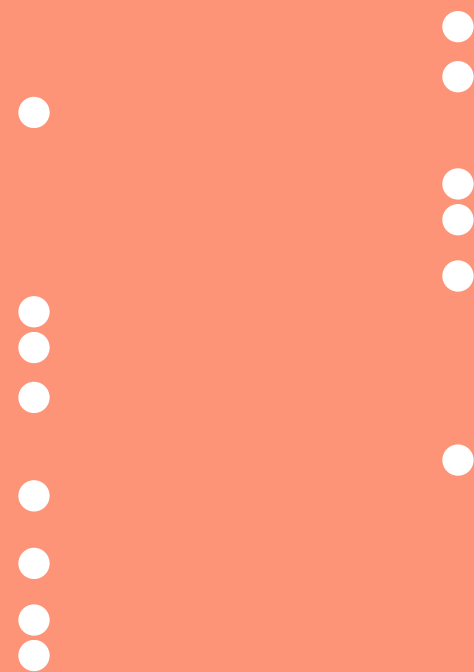
- Chandola, et al. (2009)

Types of anomalies

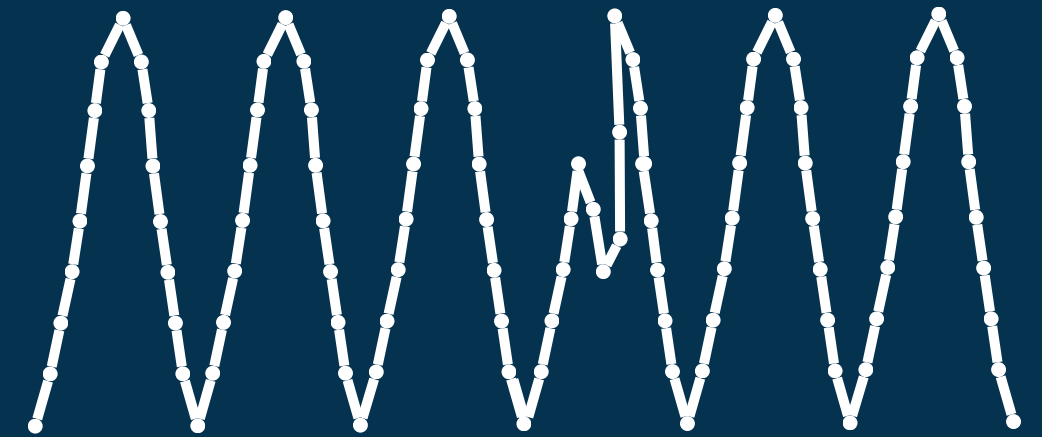
Point Anomalies

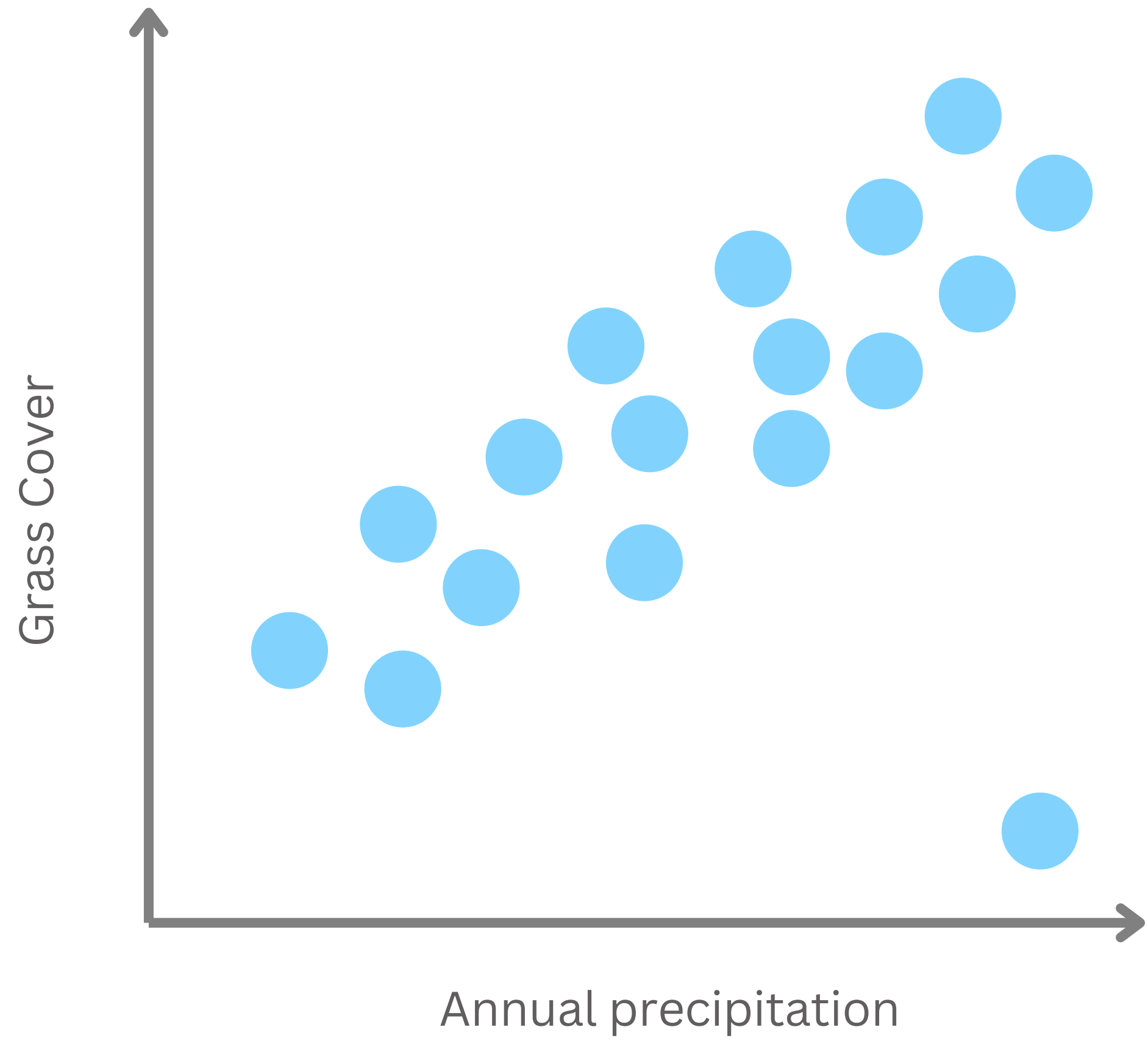


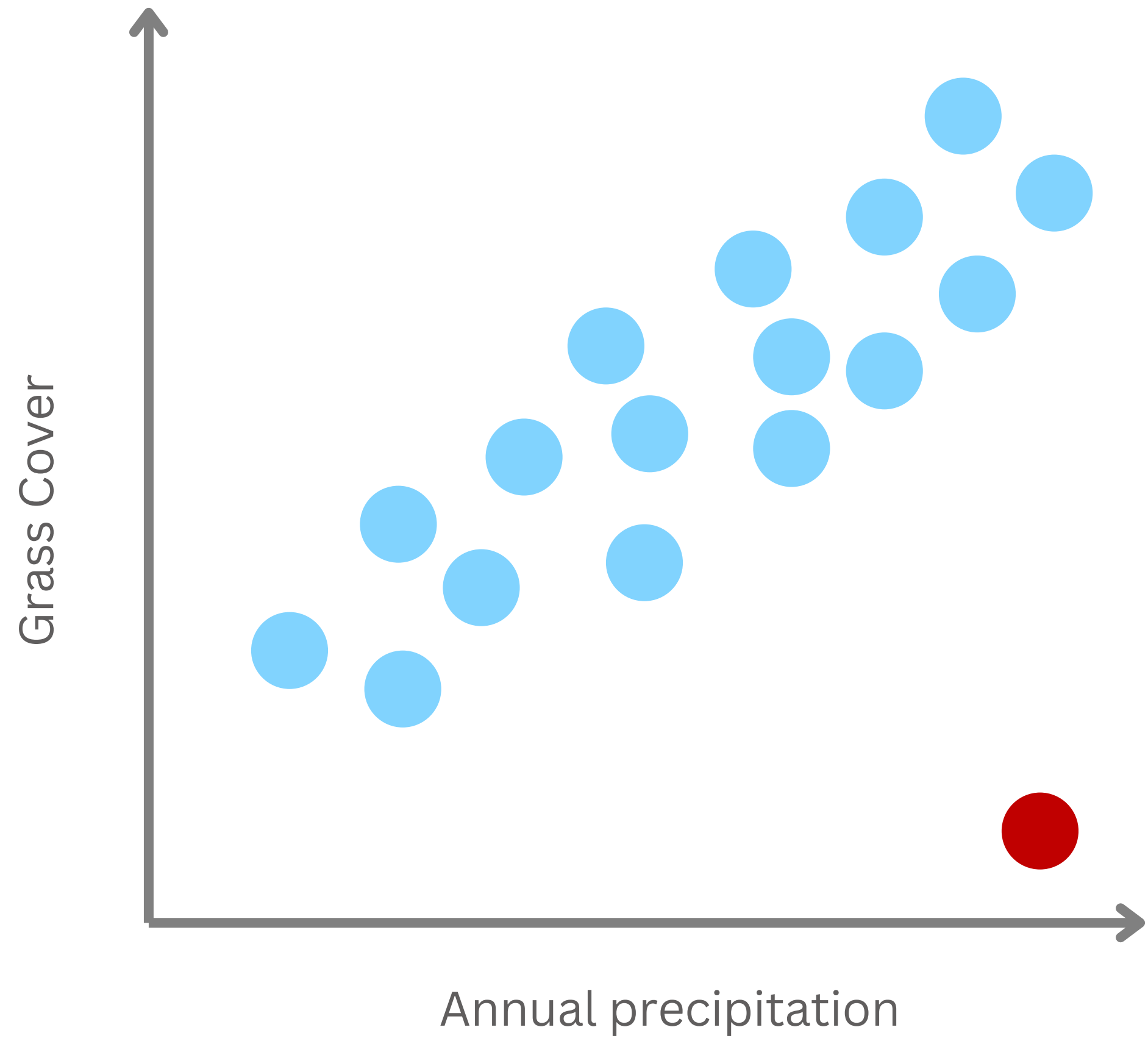
Contextual Anomalies

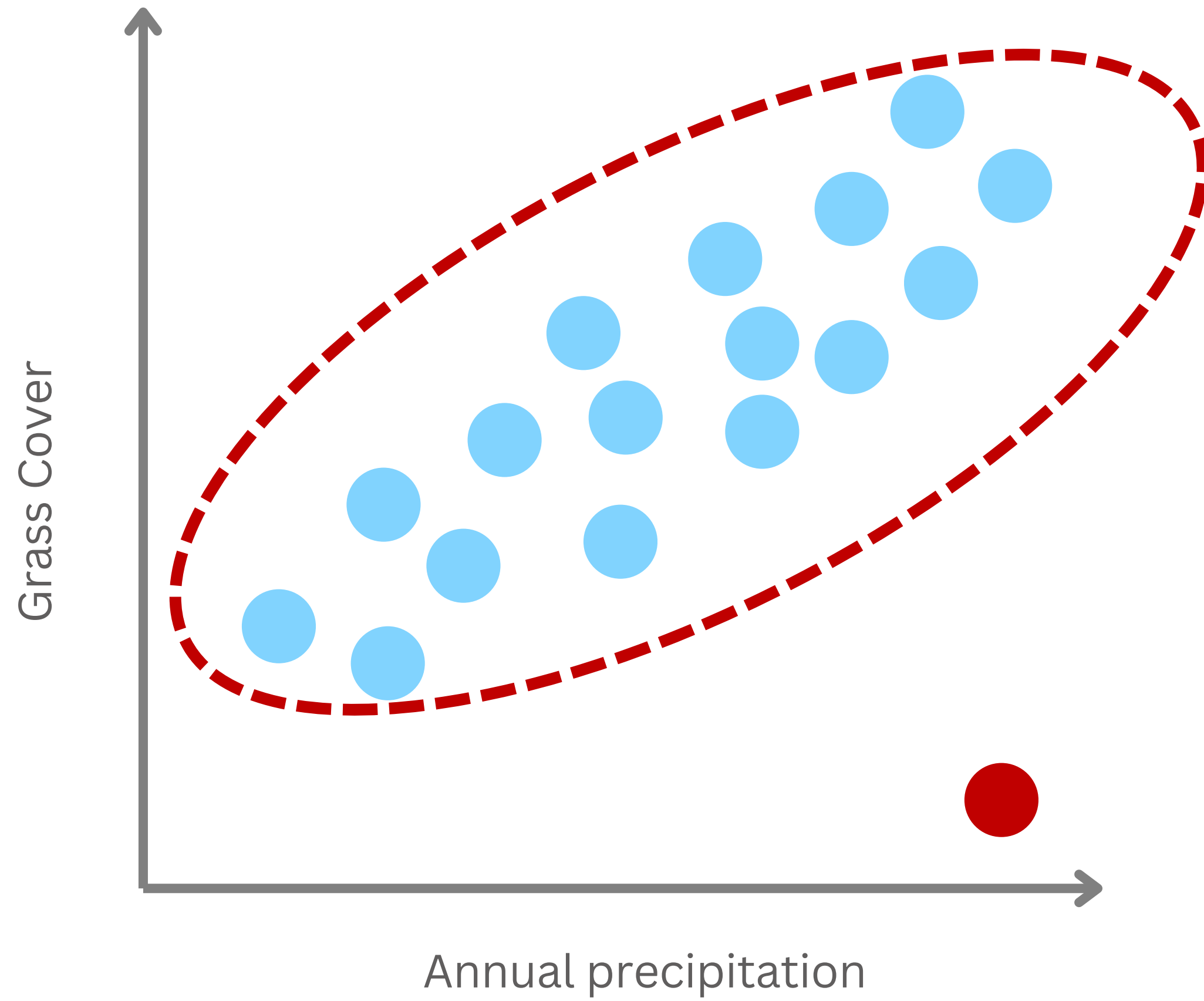


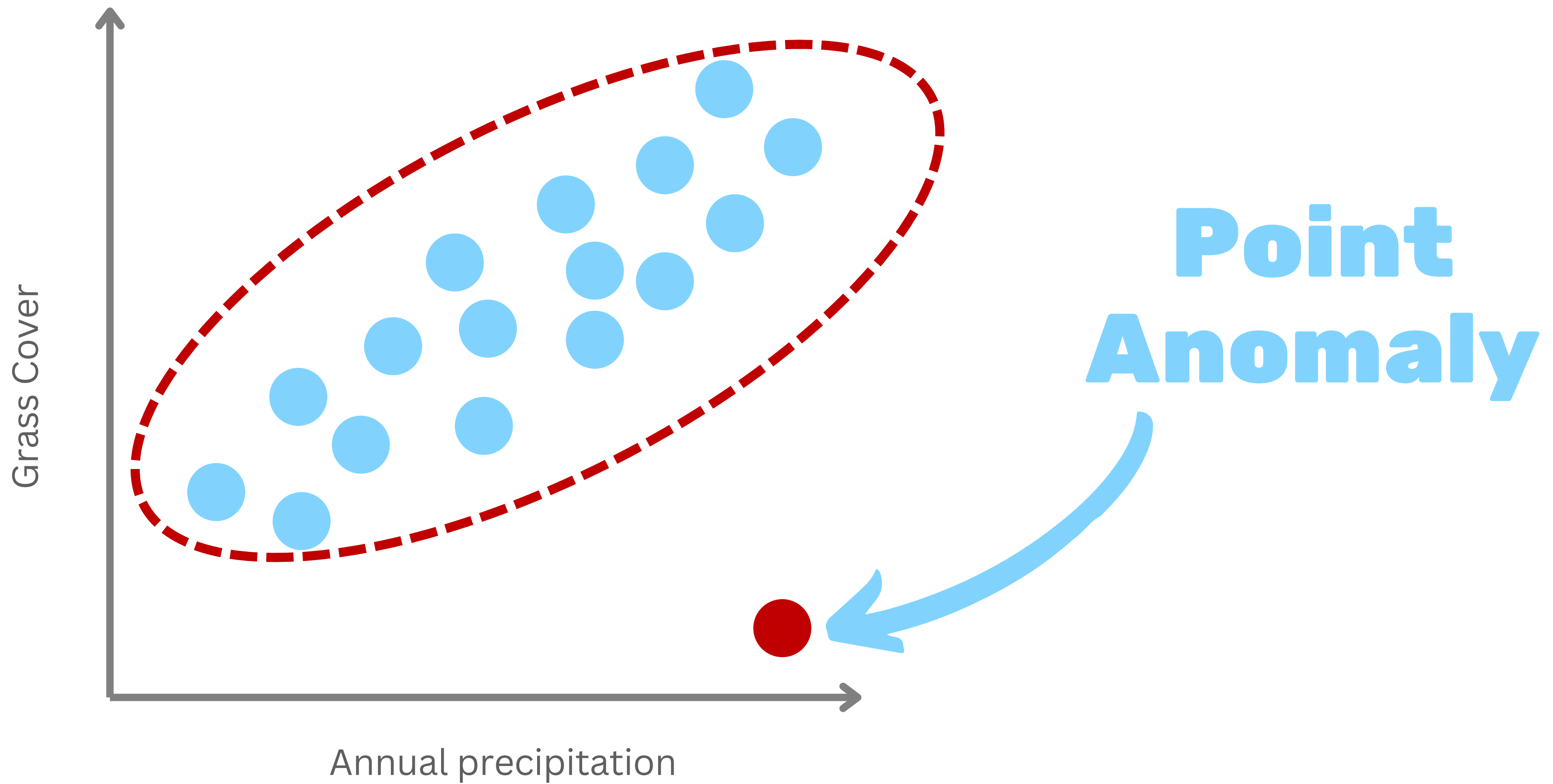
Collective Anomalies

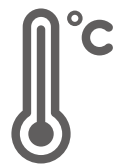












°C

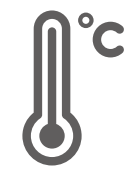
temperature

30°C

15°C

0°C





°C

temperature

30°C

15°C

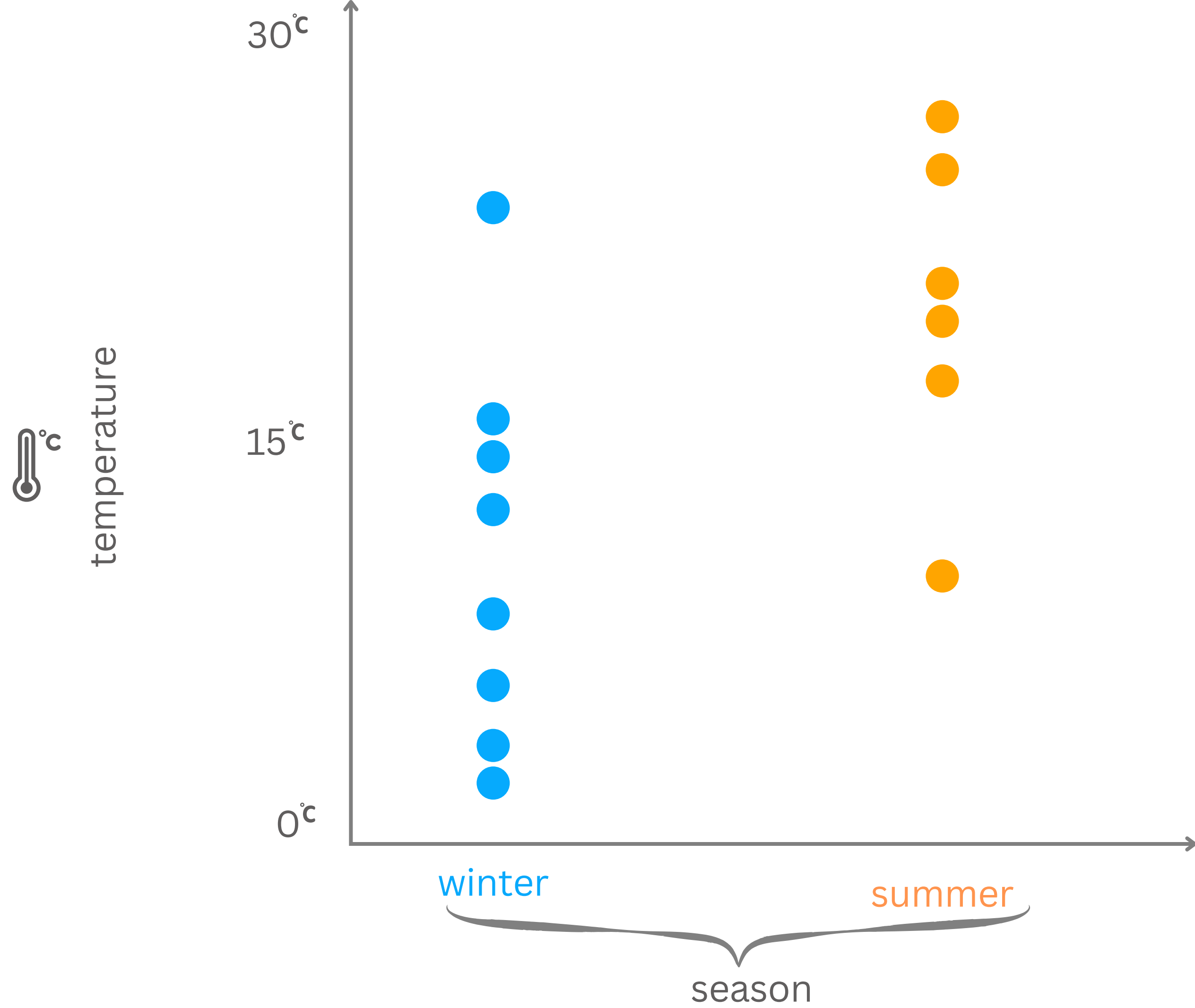
0°C

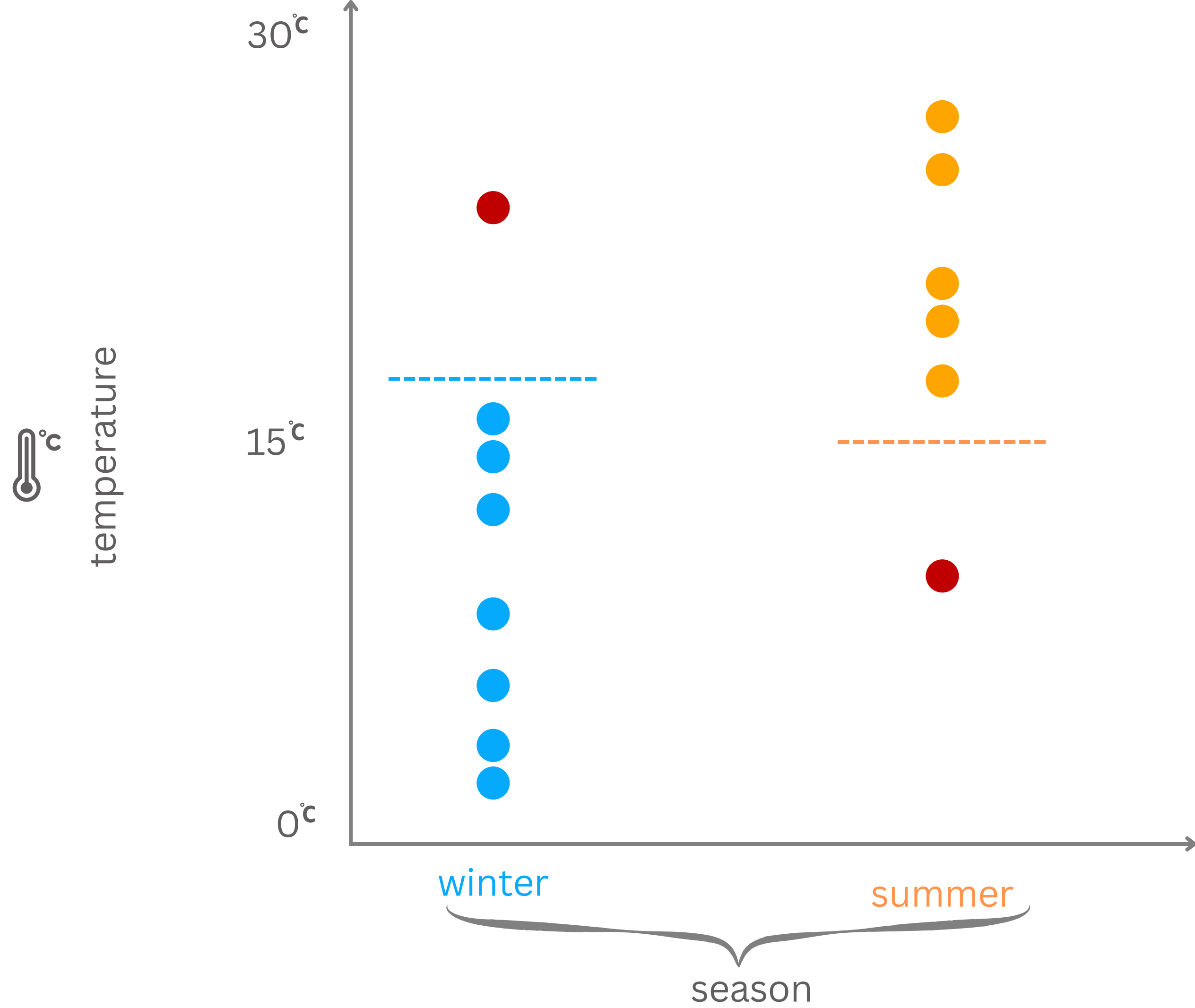


summer

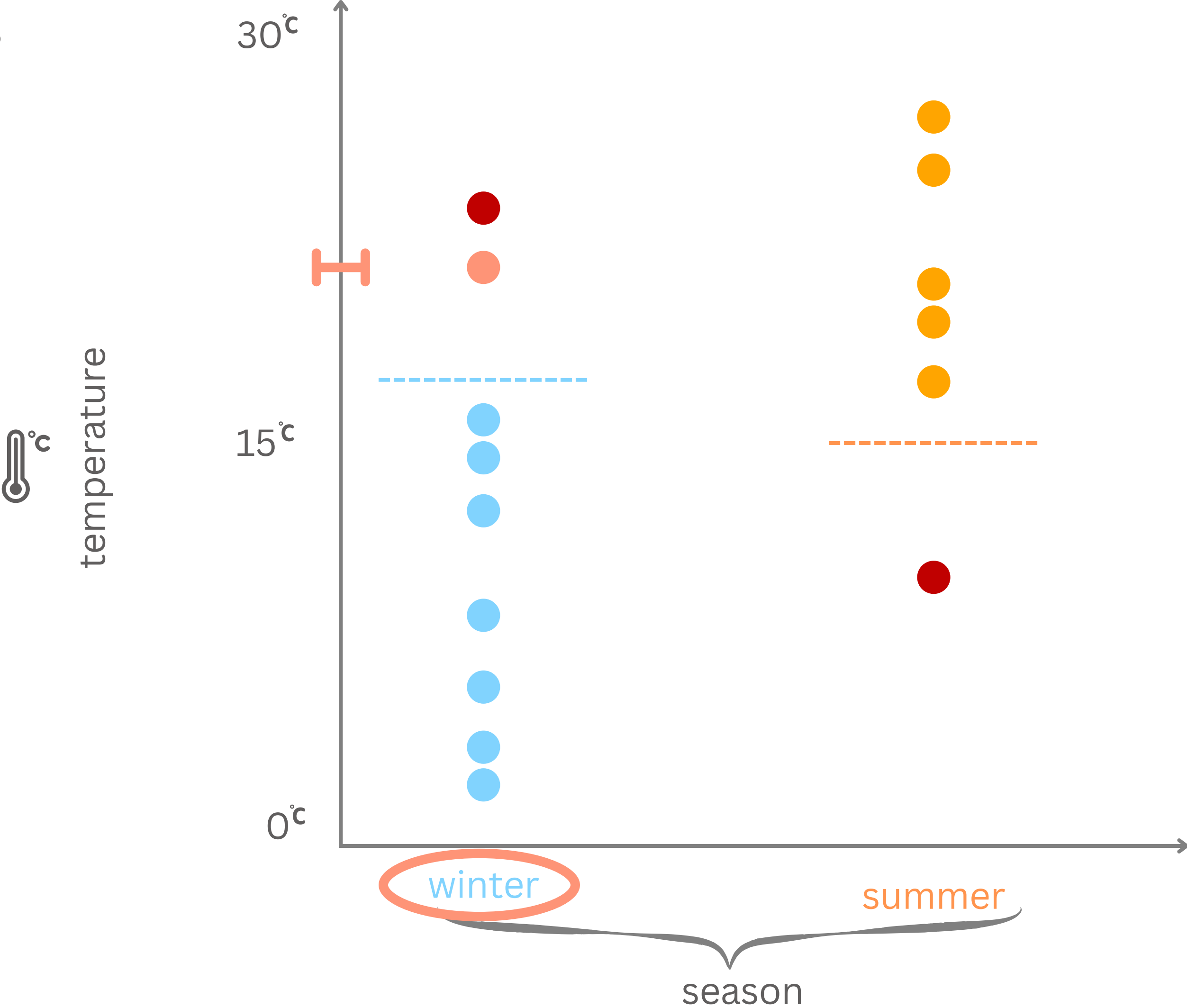


winter





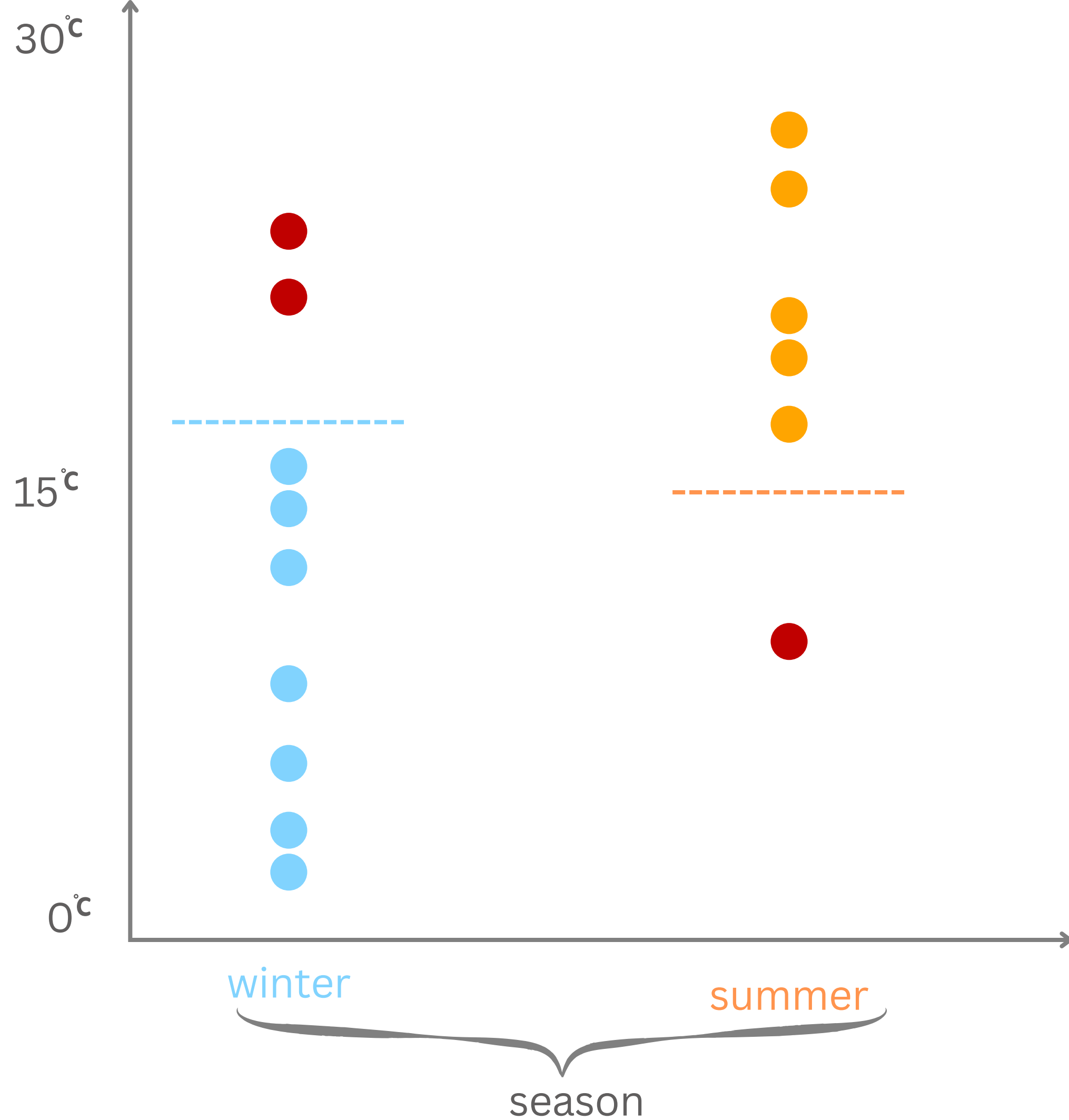
IF season=winter AND temperature > 16



IF season=**winter** AND temperature > 16
THEN datapoint is an **outlier**



temperature

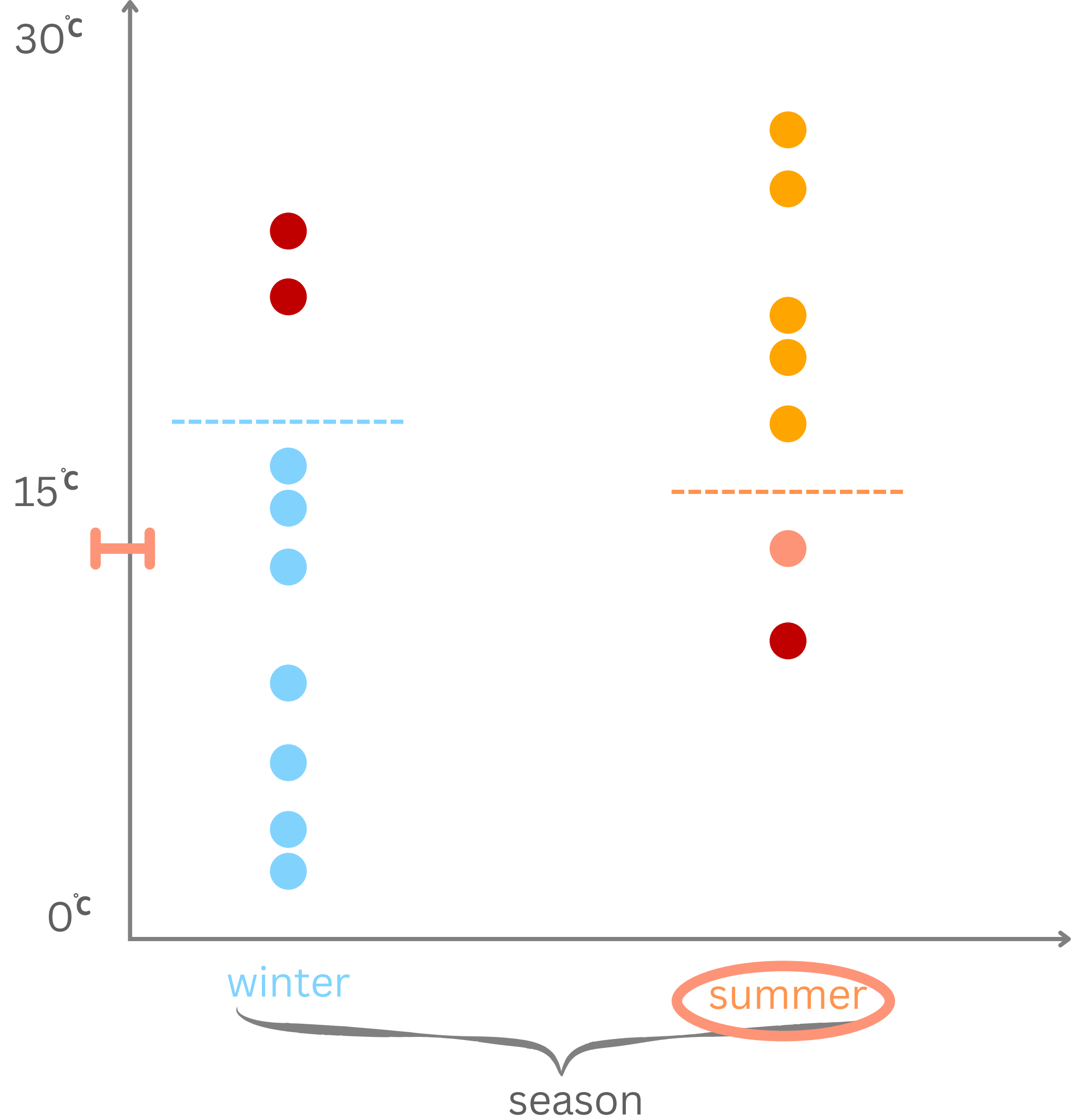


IF season=**winter** AND temperature > 16
THEN datapoint is an **outlier**

ELSE IF season=**summer** AND temperature < 15



temperature

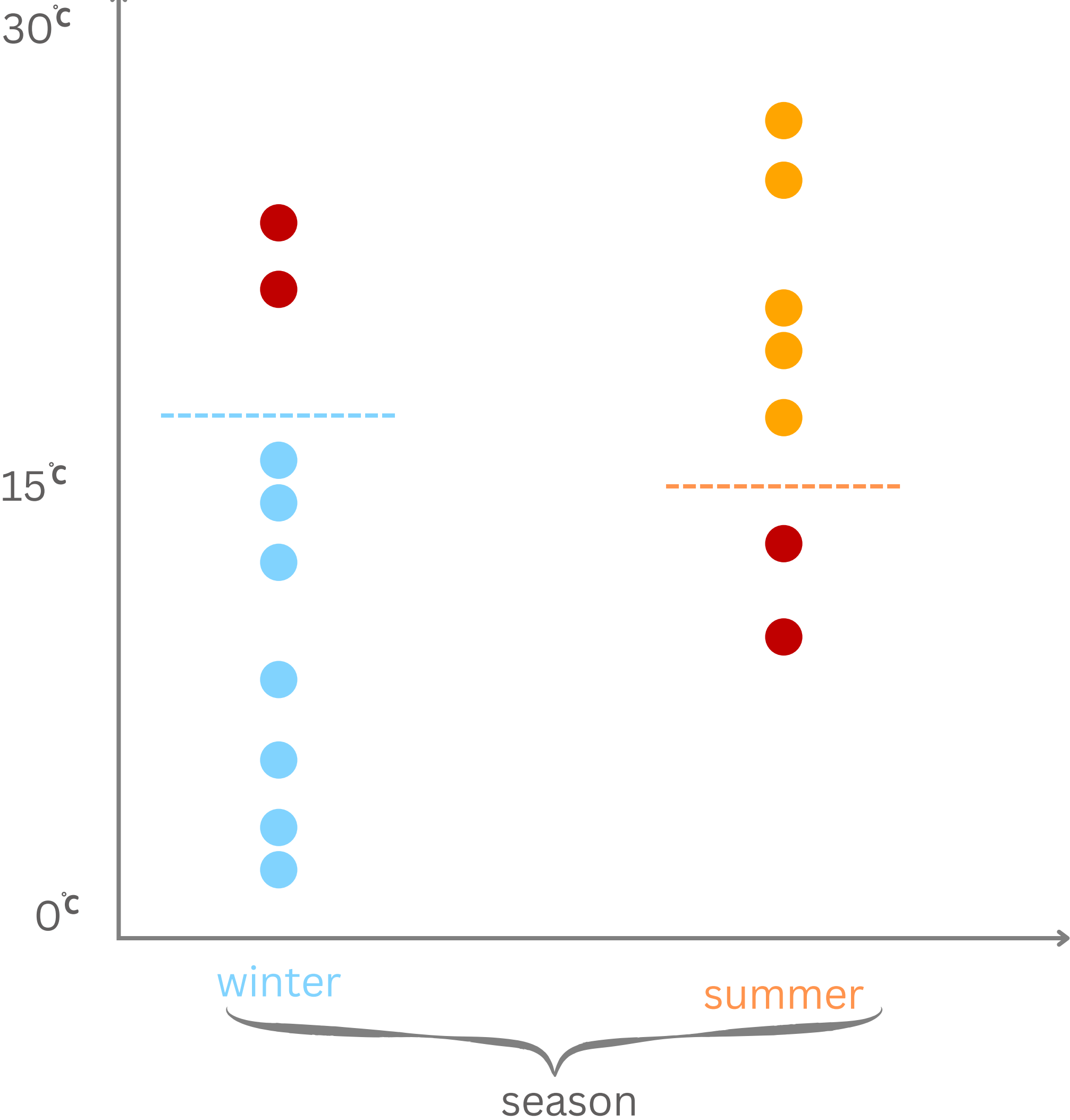


IF season=**winter** AND temperature > 16
THEN datapoint is an **outlier**

ELSE IF season=**summer** AND temperature < 15
THEN datapoint is an **outlier**



temperature



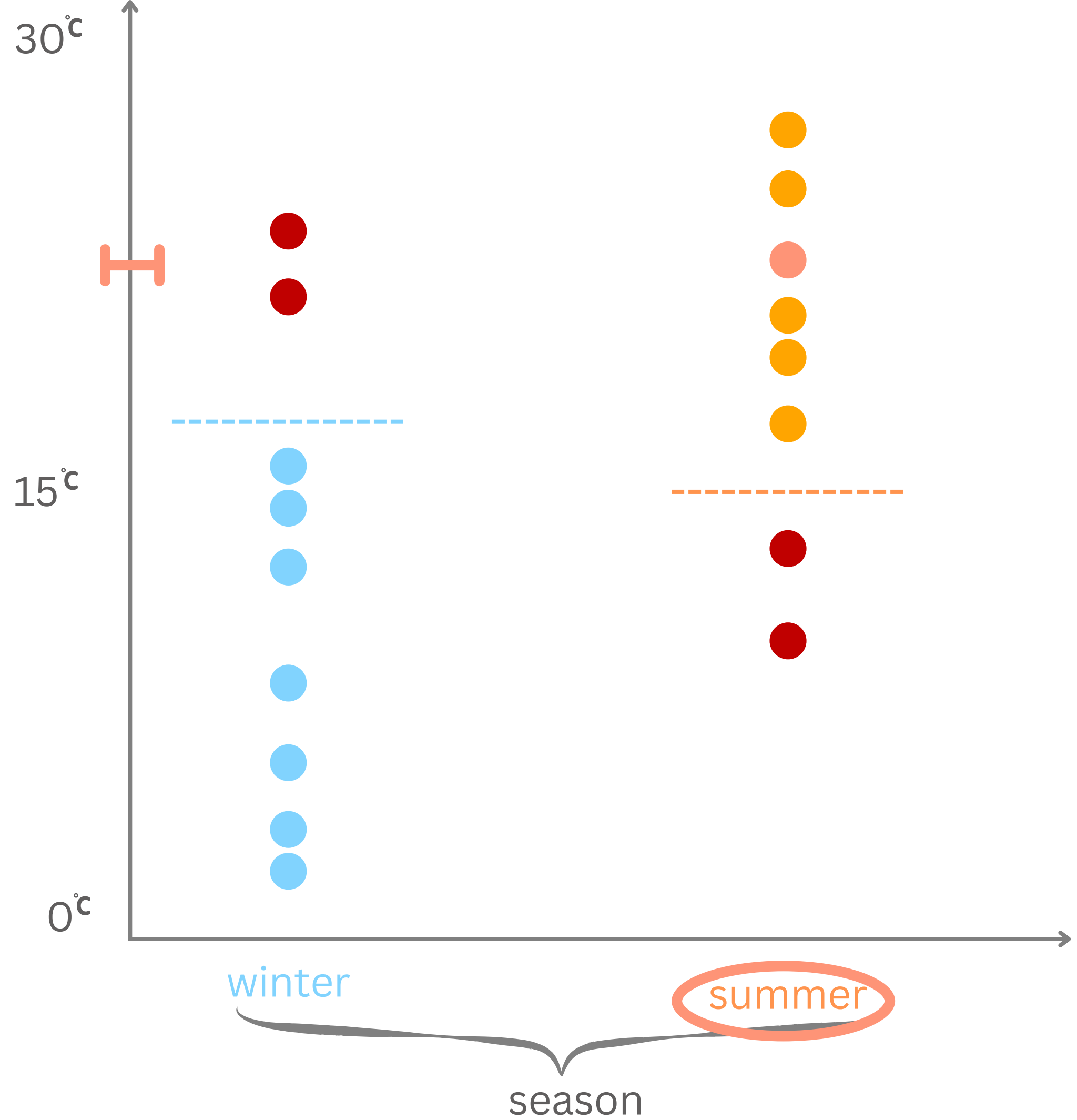
IF season=**winter** AND temperature > 16
THEN datapoint is an **outlier**

ELSE IF season=**summer** AND temperature < 15
THEN datapoint is an **outlier**

ELSE



temperature



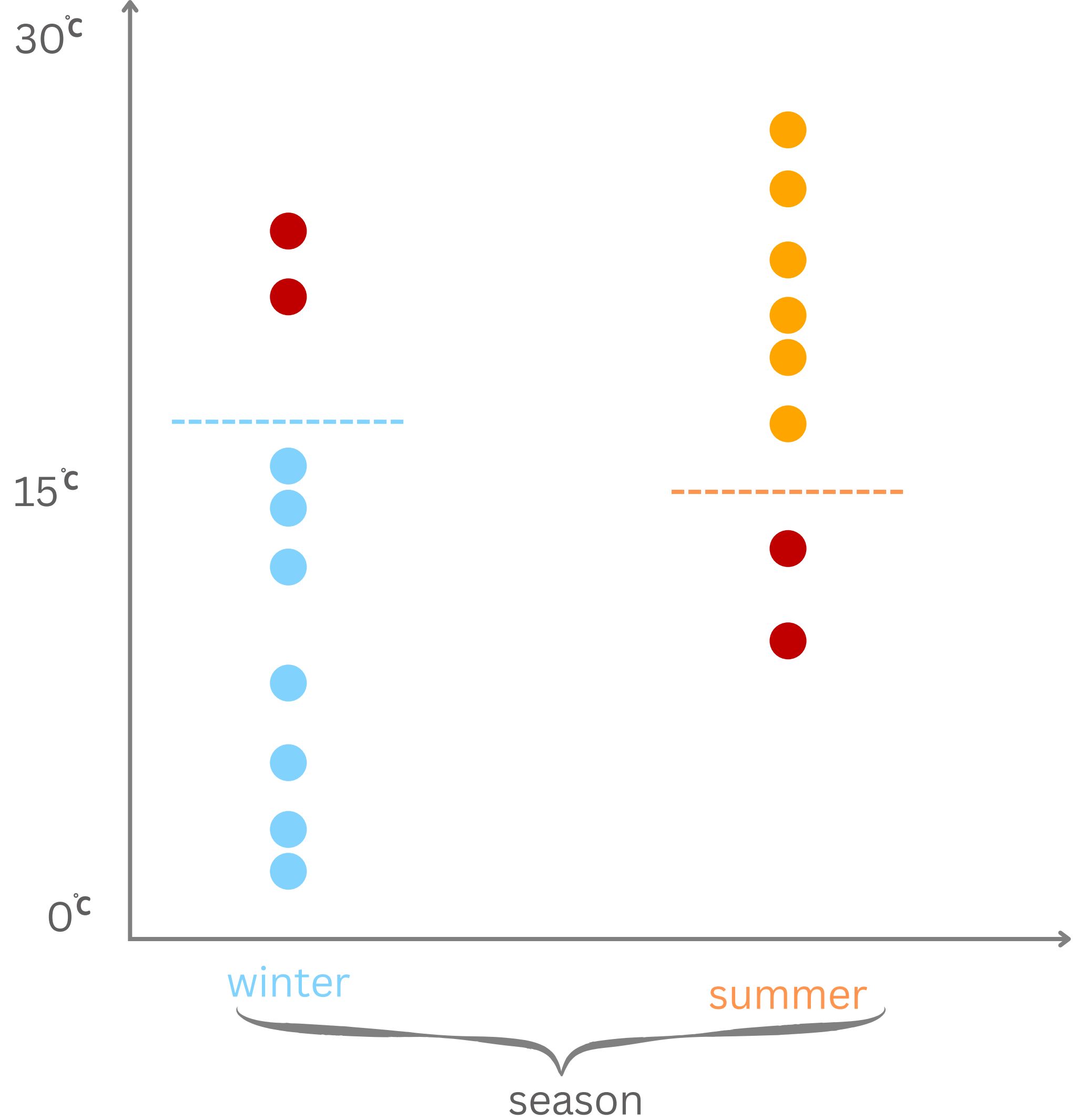
IF season=**winter** AND temperature > 16
THEN datapoint is an **outlier**

ELSE IF season=**summer** AND temperature < 15
THEN datapoint is an **outlier**

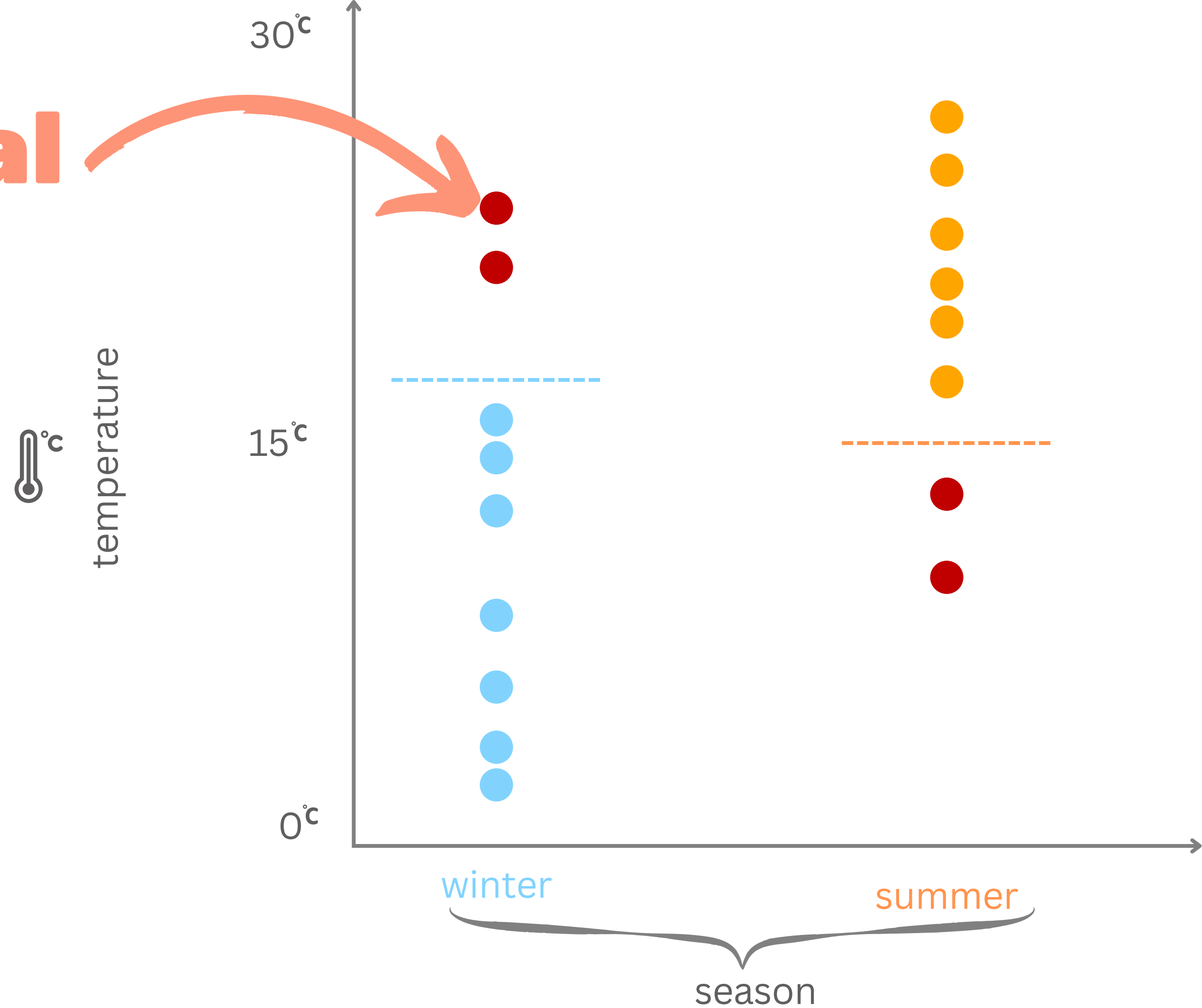
ELSE
datapoint is **normal**

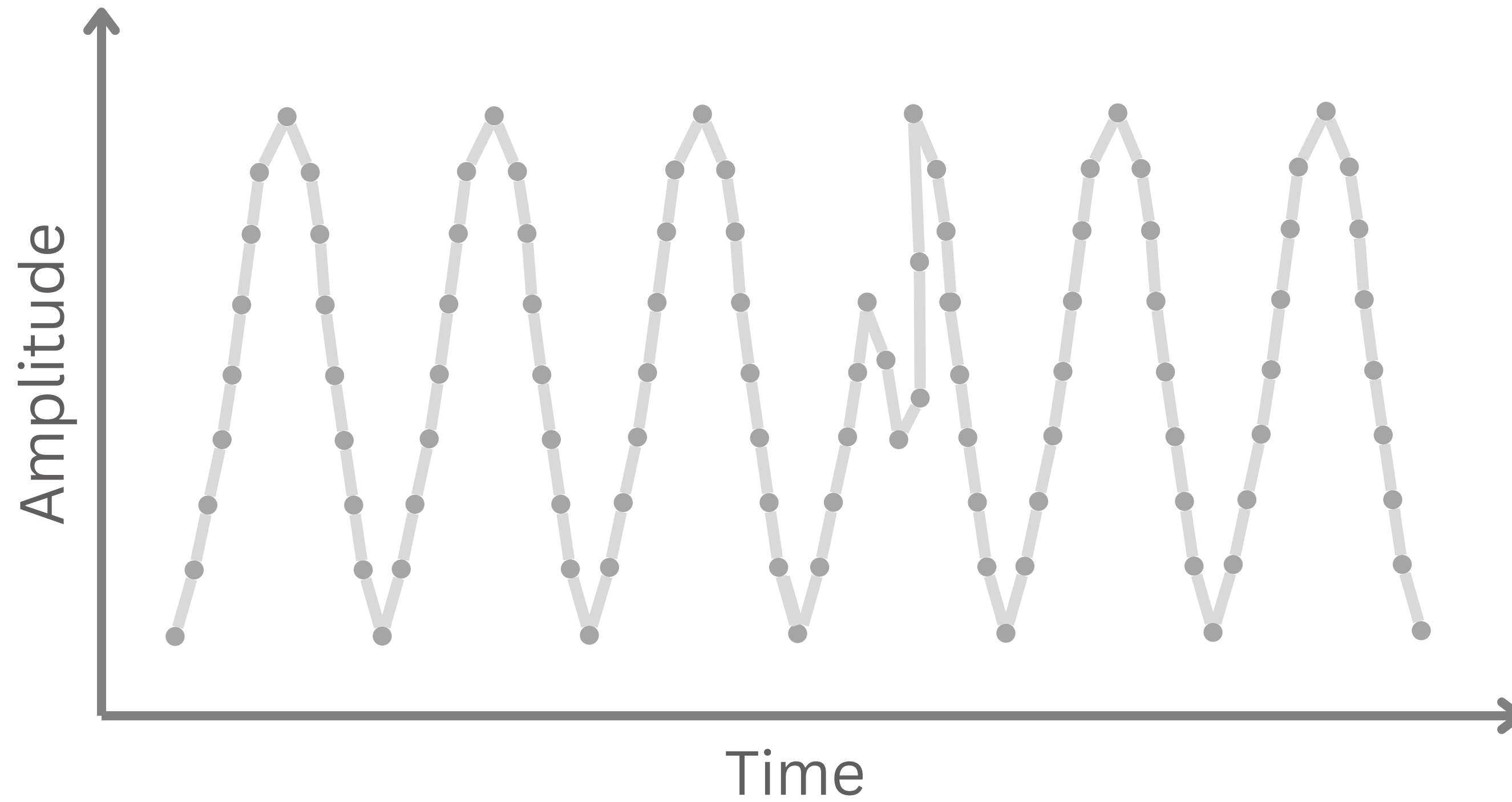


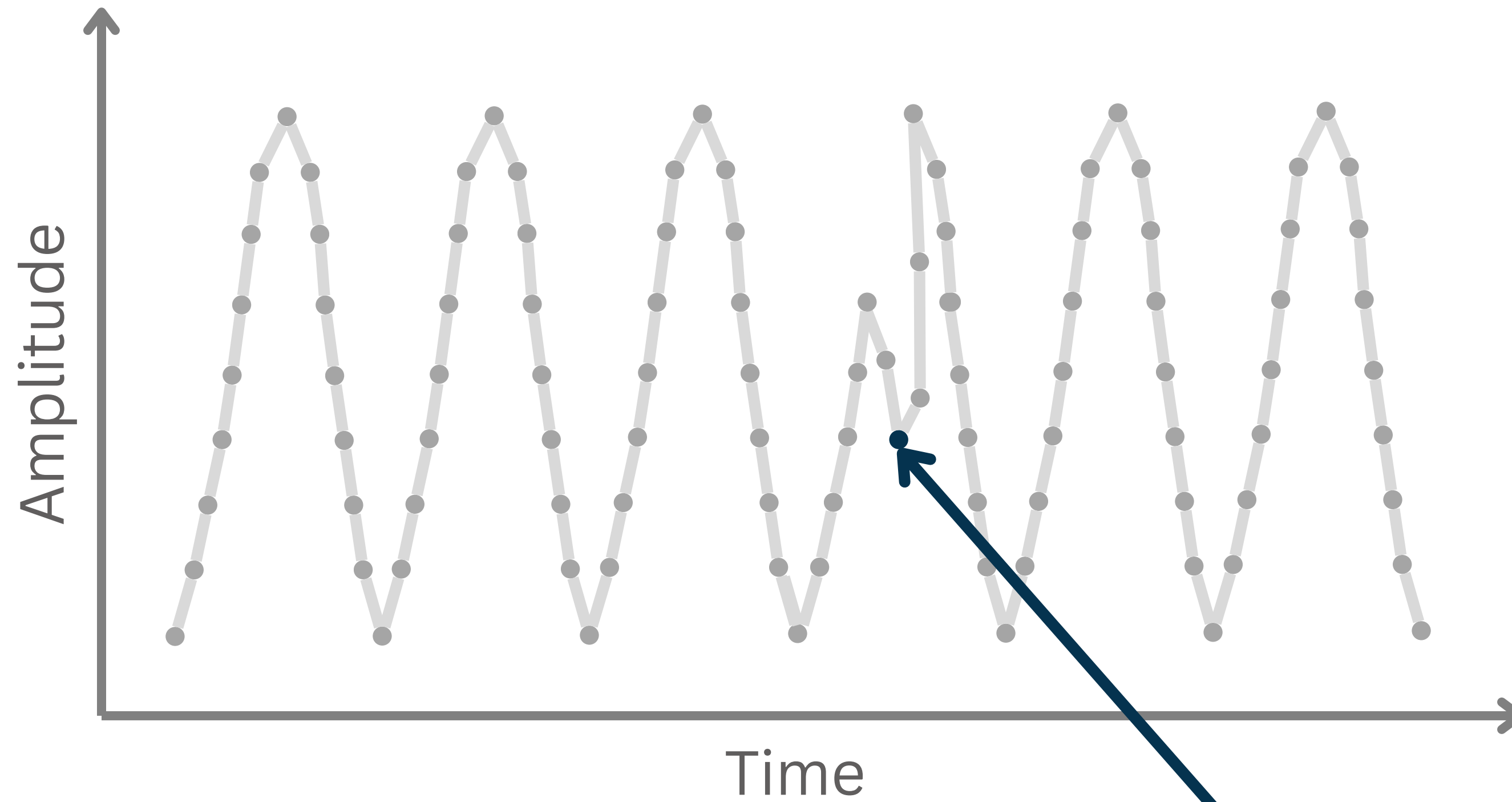
temperature



Contextual Anomaly

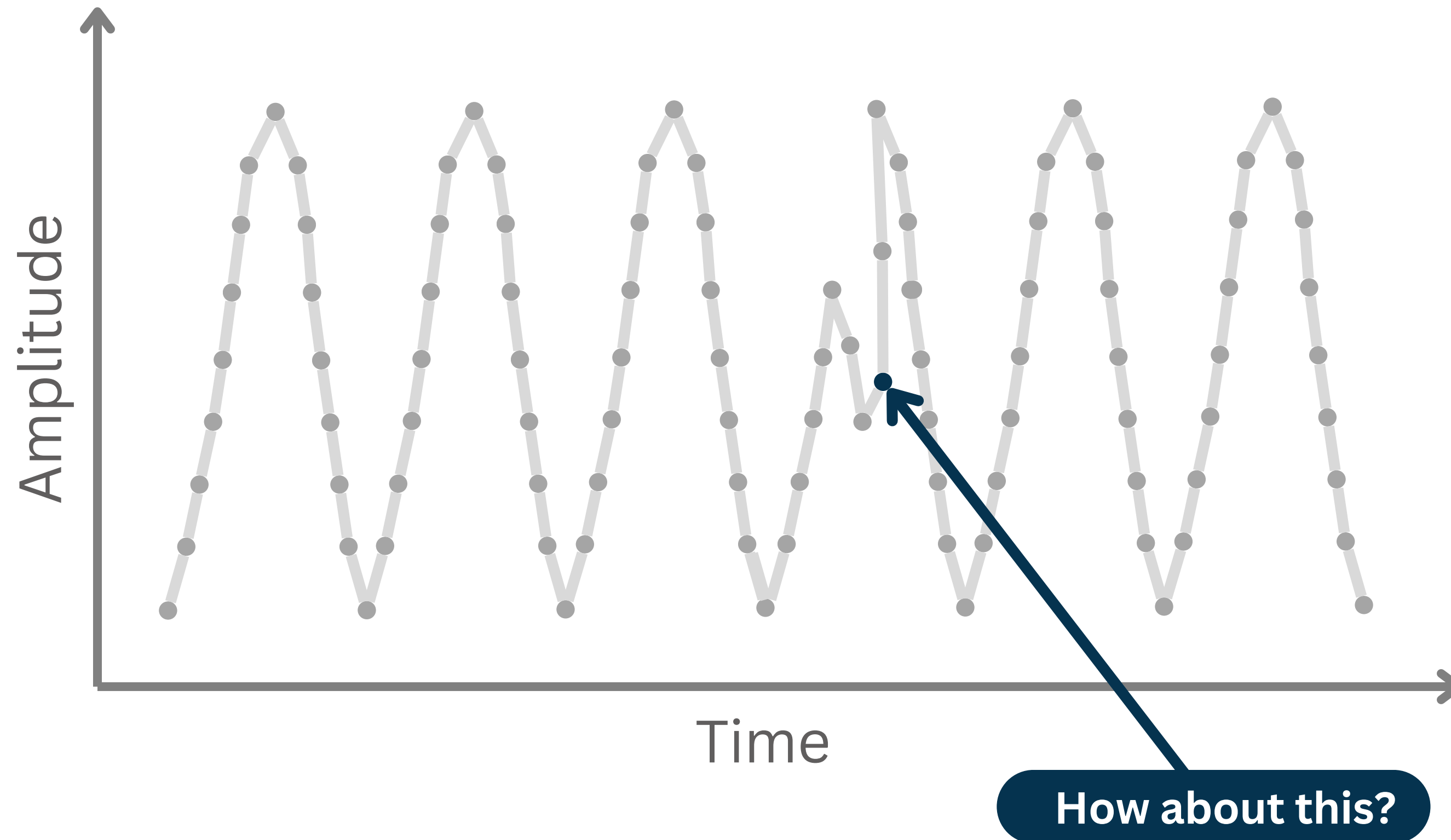


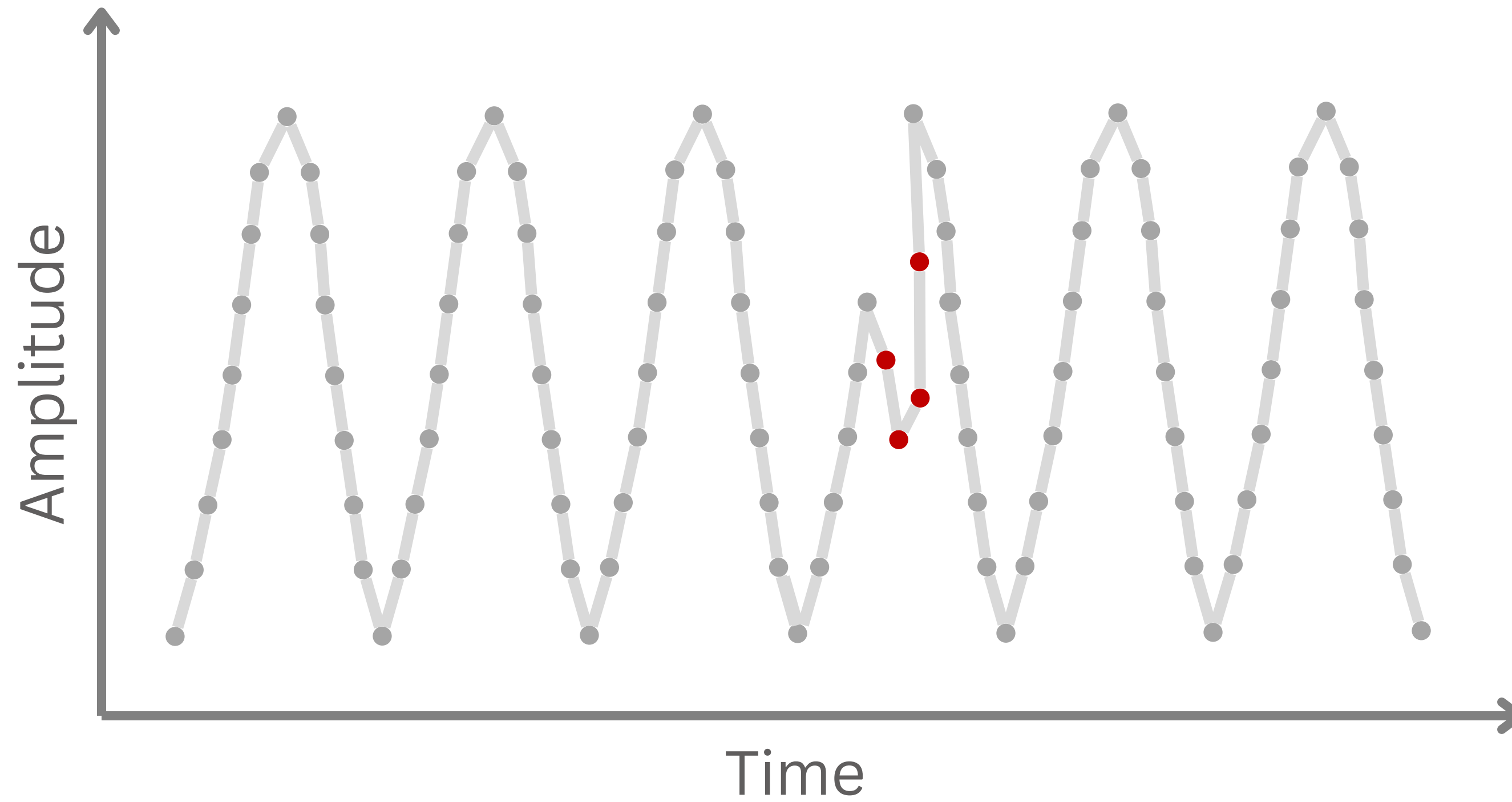




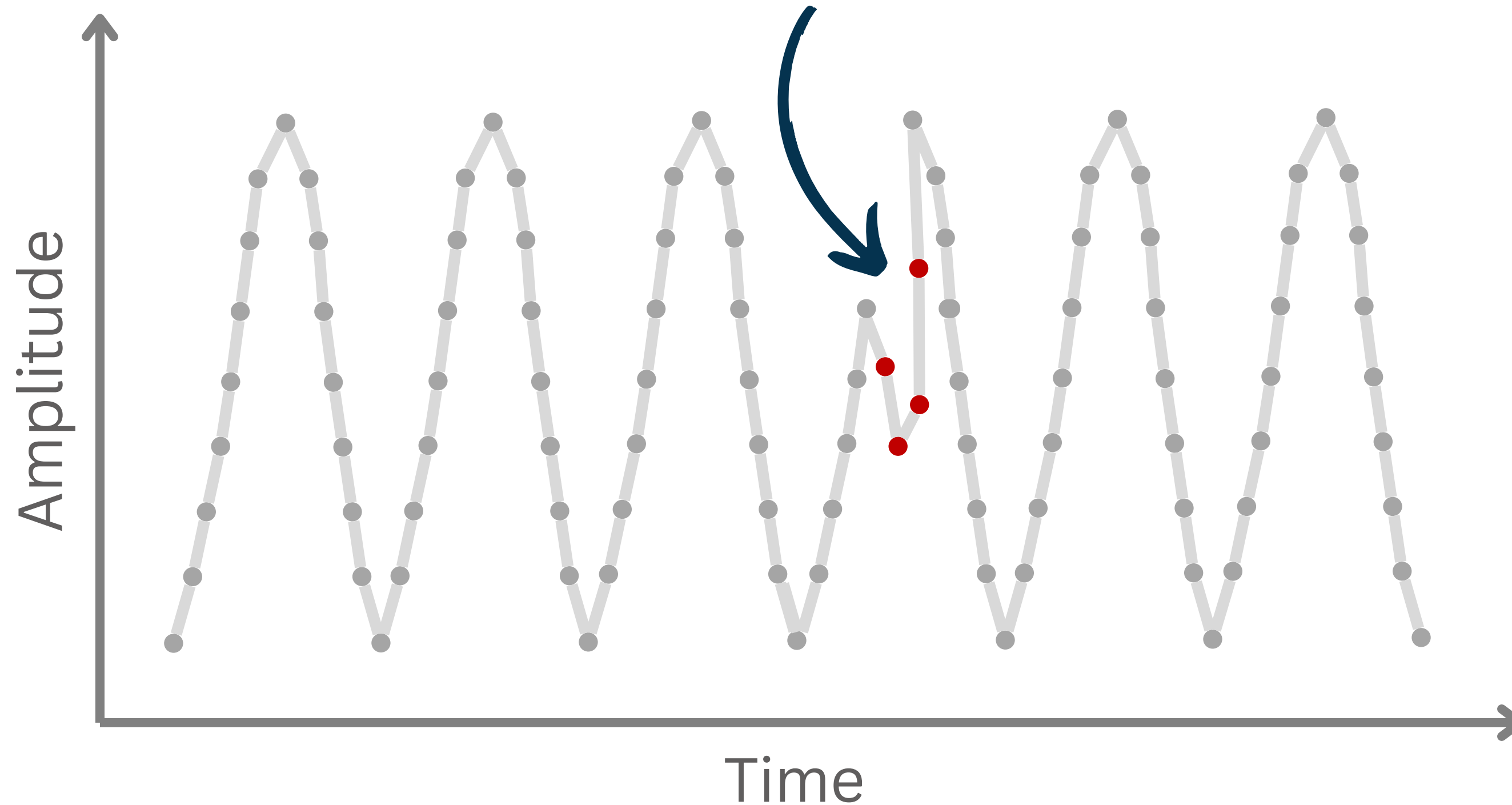
Is this an anomaly?

If so, why?

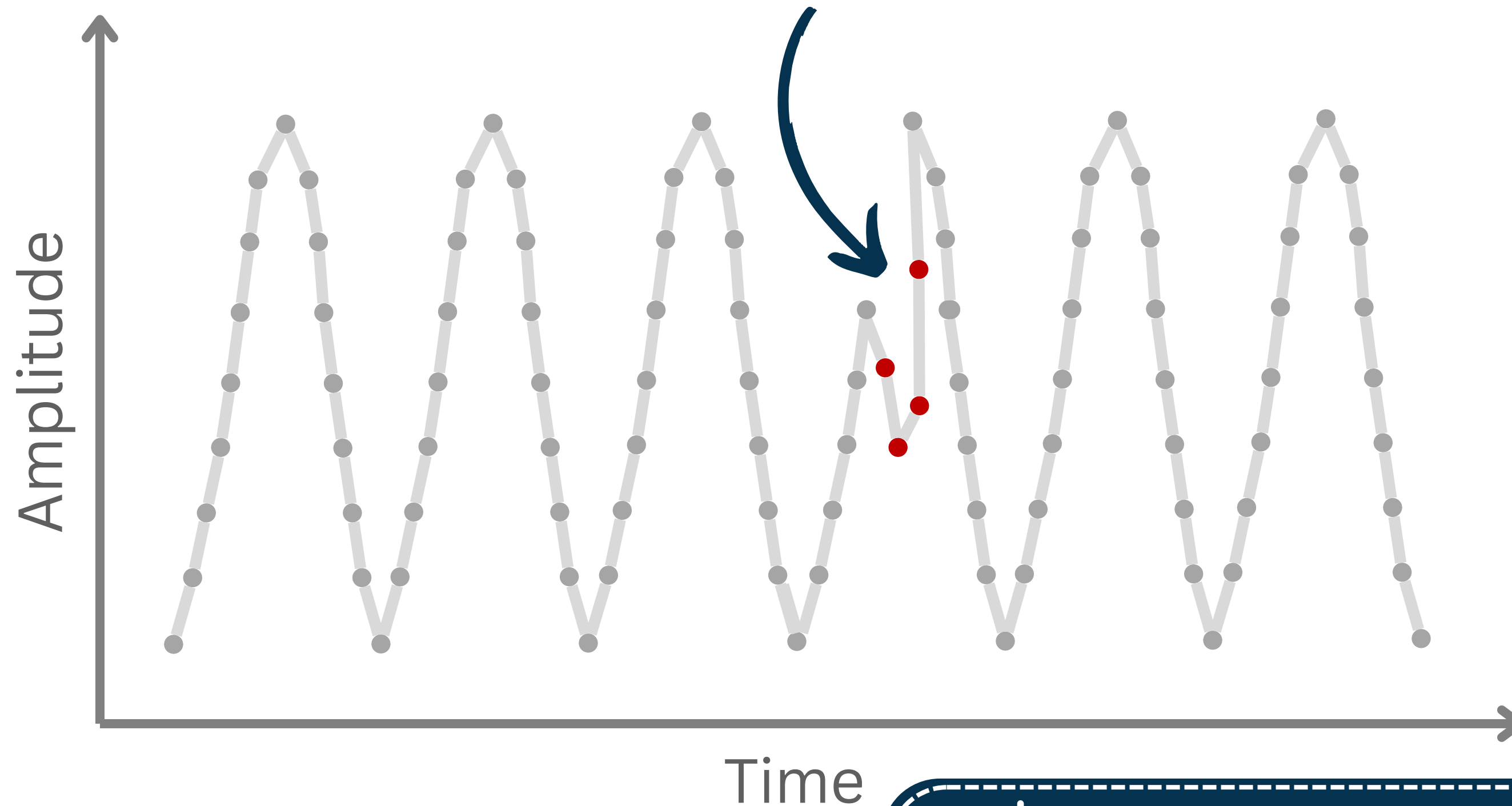




Collective Anomaly



Collective Anomaly

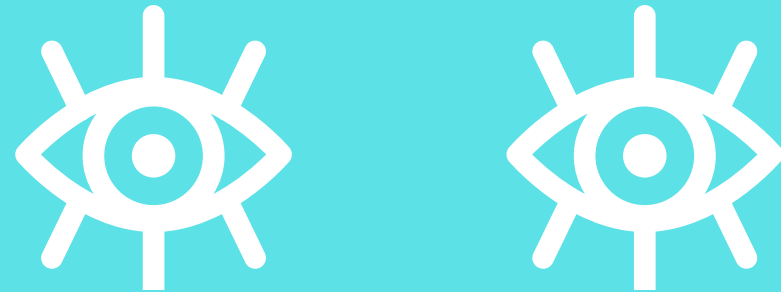


Anomalies are decided
by more complex rules
or functions

**How do we
approach anomaly
detection?**

Approaches

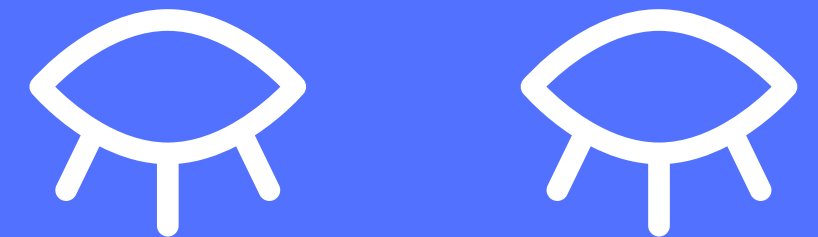
Supervised?



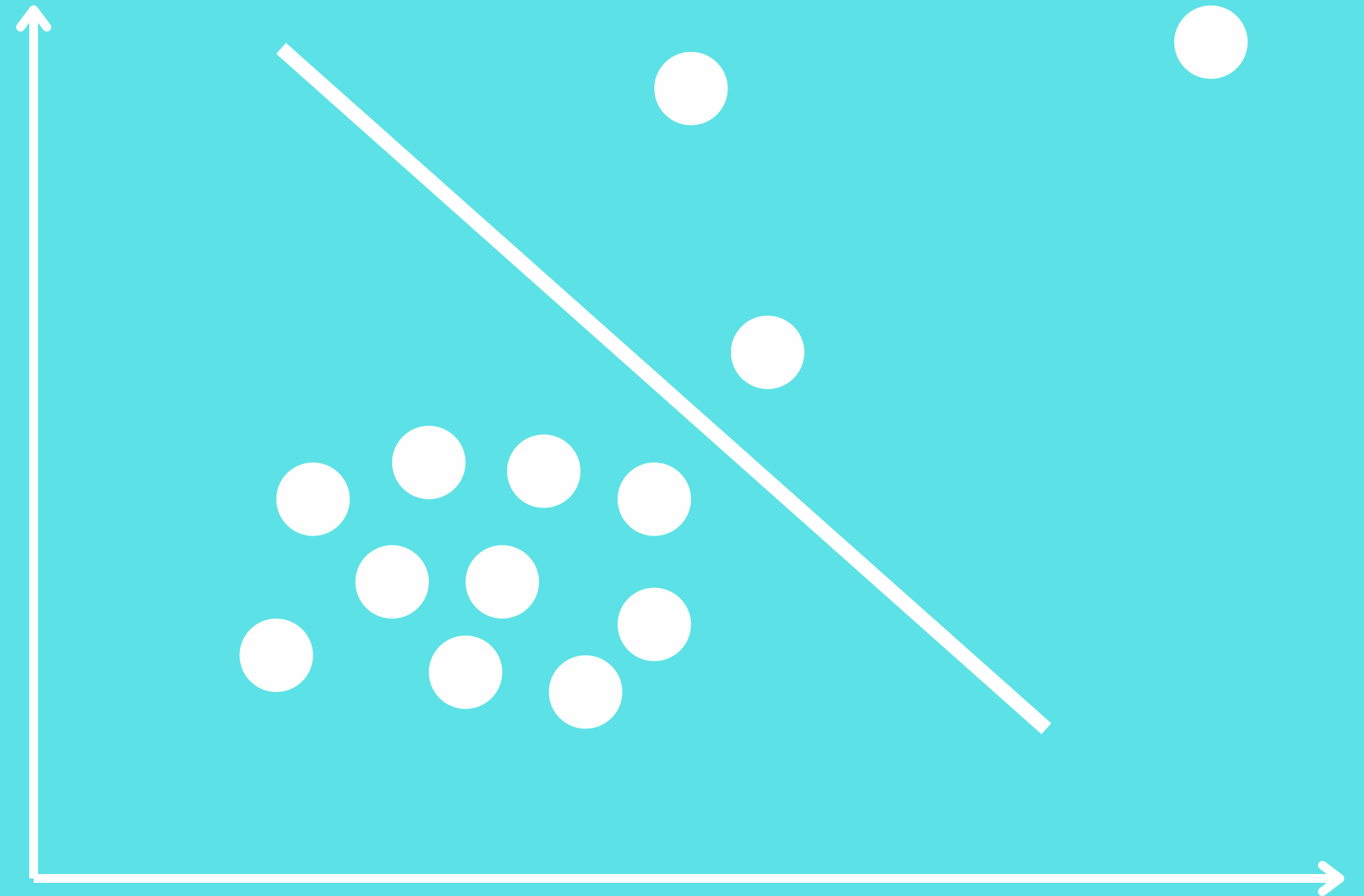
Semi-supervised



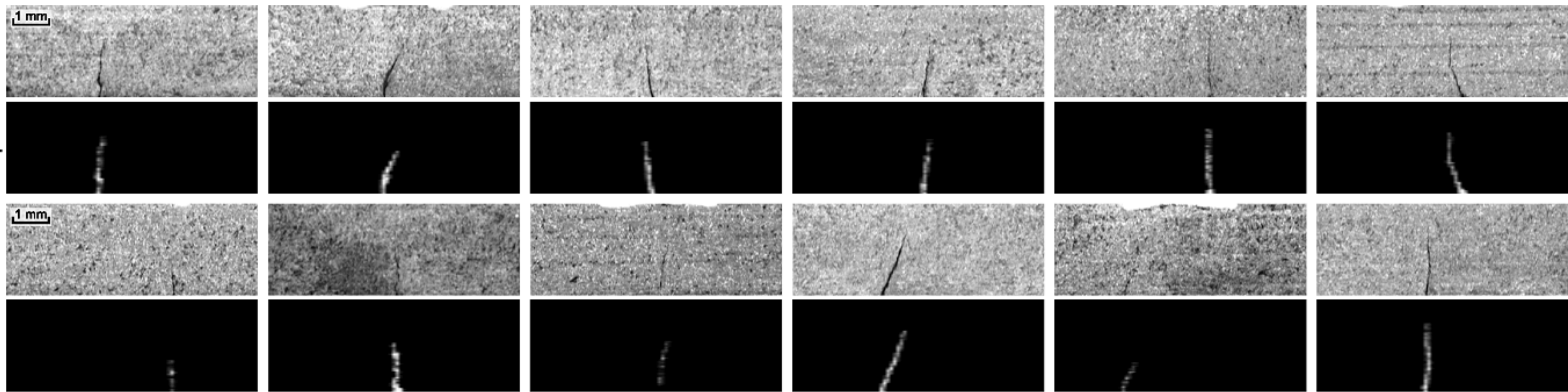
Unsupervised



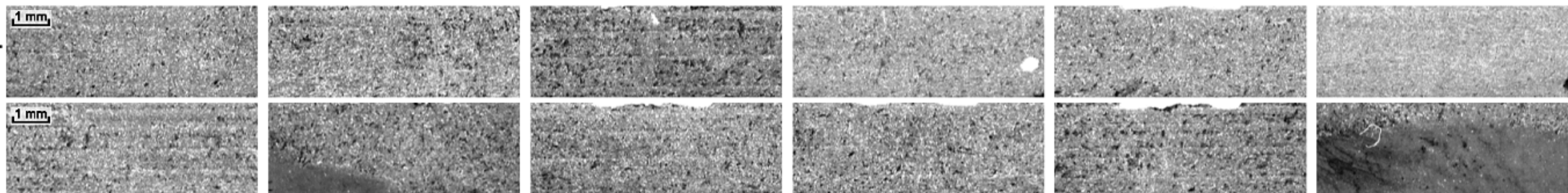
SUPERVISED ANOMALY DETECTION



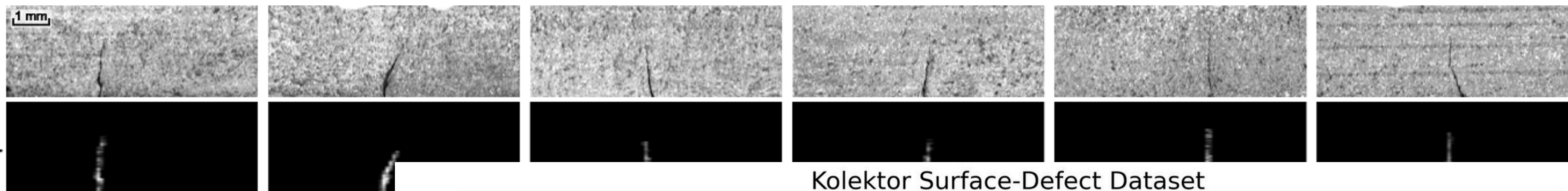
Defective samples



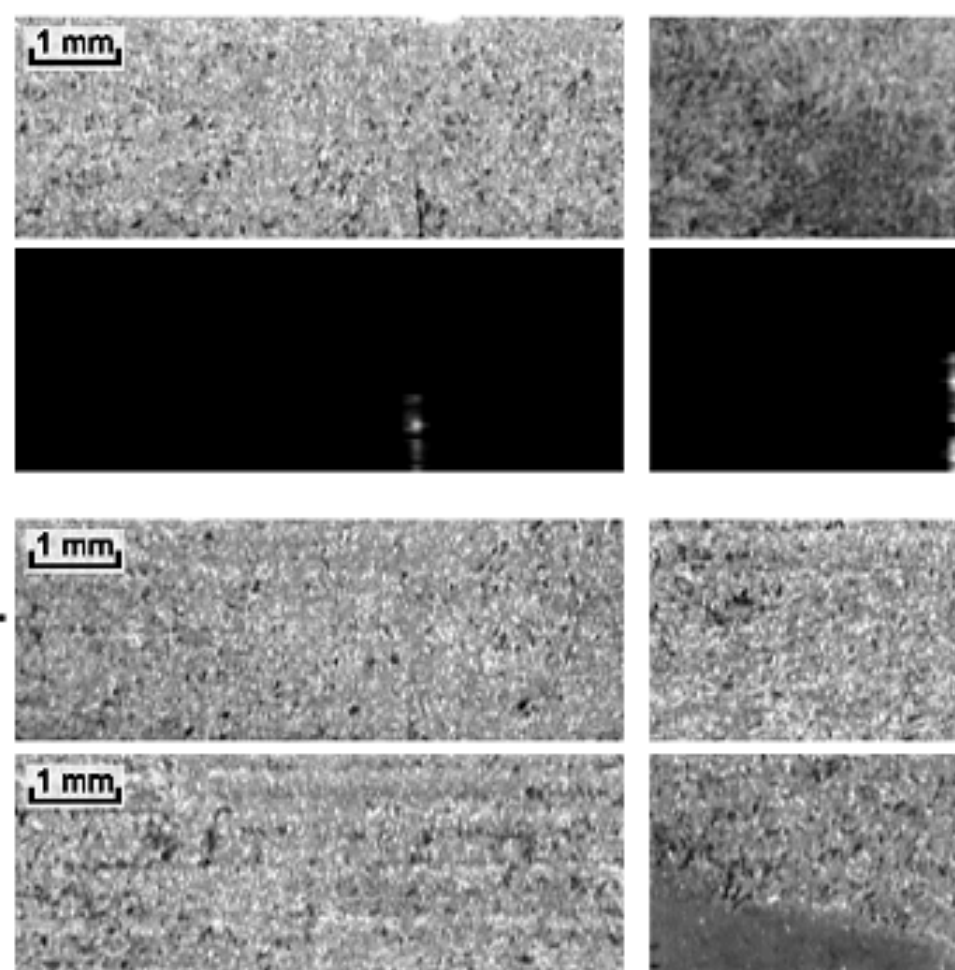
Defect-free samples



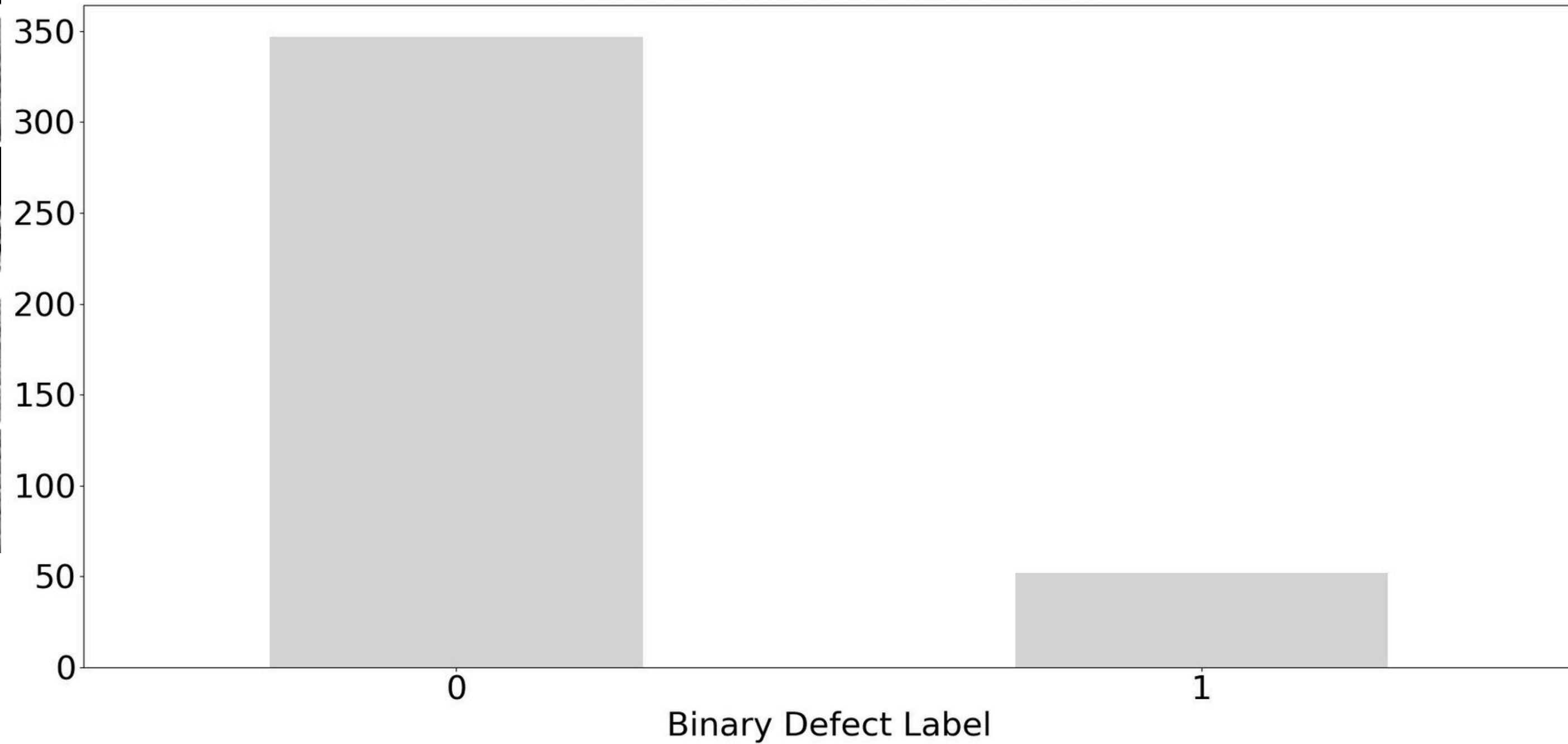
Defective samples



Defect-free samples



Kolektor Surface-Defect Dataset





Thyroid Disease

Donated on 12/31/1986

10 separate databases from Garavan Institute

Dataset Characteristics

Multivariate, Domain-Theory

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Categorical, Real

Instances

7200

Features

5

Dataset Information

Additional Information

From Garavan Institute

Documentation: as given by Ross Quinlan

6 databases from the Garavan Institute in Sydney, Australia

Approximately the following for each database:

DOWNLOAD

CITE

” 18 citations

👁 104213 views

Creators

👤 Ross Quinlan

DOI

10.24432/C5D010

License

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Thyroid Disease

Donated on 12/31/1986

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18 citations

10 separate databases from Garavan Institute

Dataset Characteristics

Multivariate, Domain-Theory

Feature Type

Categorical, Real

Dataset Information

Additional Information

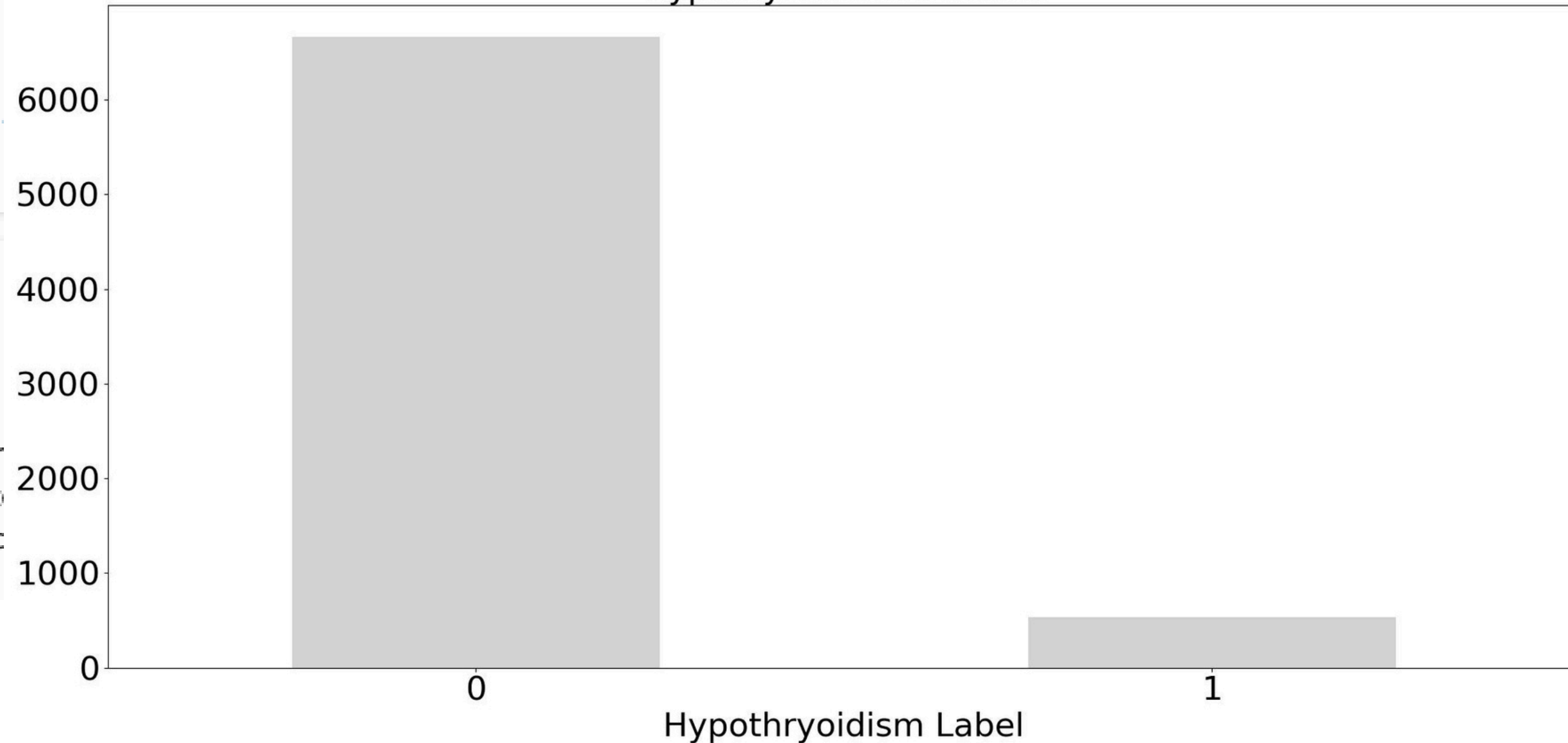
From Garavan Institute

Documentation: as given by Ross Quir

6 databases from the Garavan Institut

Approximately the following for each c

Hypothyroidism Dataset



Anomaly Detection Datasets are

imbalanced



Sometimes

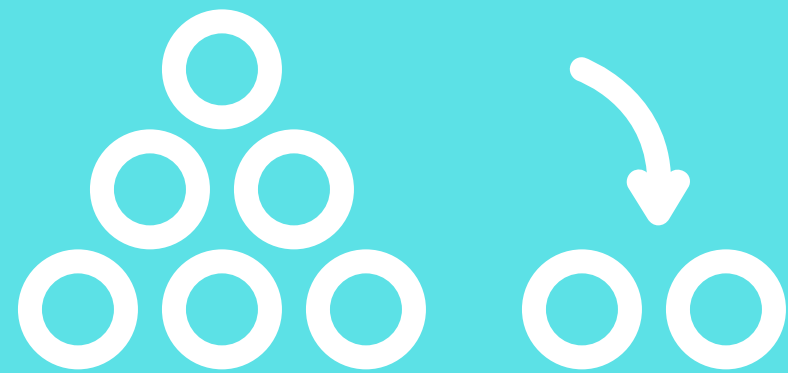
very

imbalanced

This makes binary classification **difficult**



Classifiers prone to false negatives



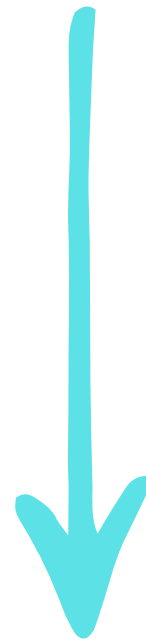
**Not enough examples to
accurately represent
different anomalies**



What are our options?



undersampling

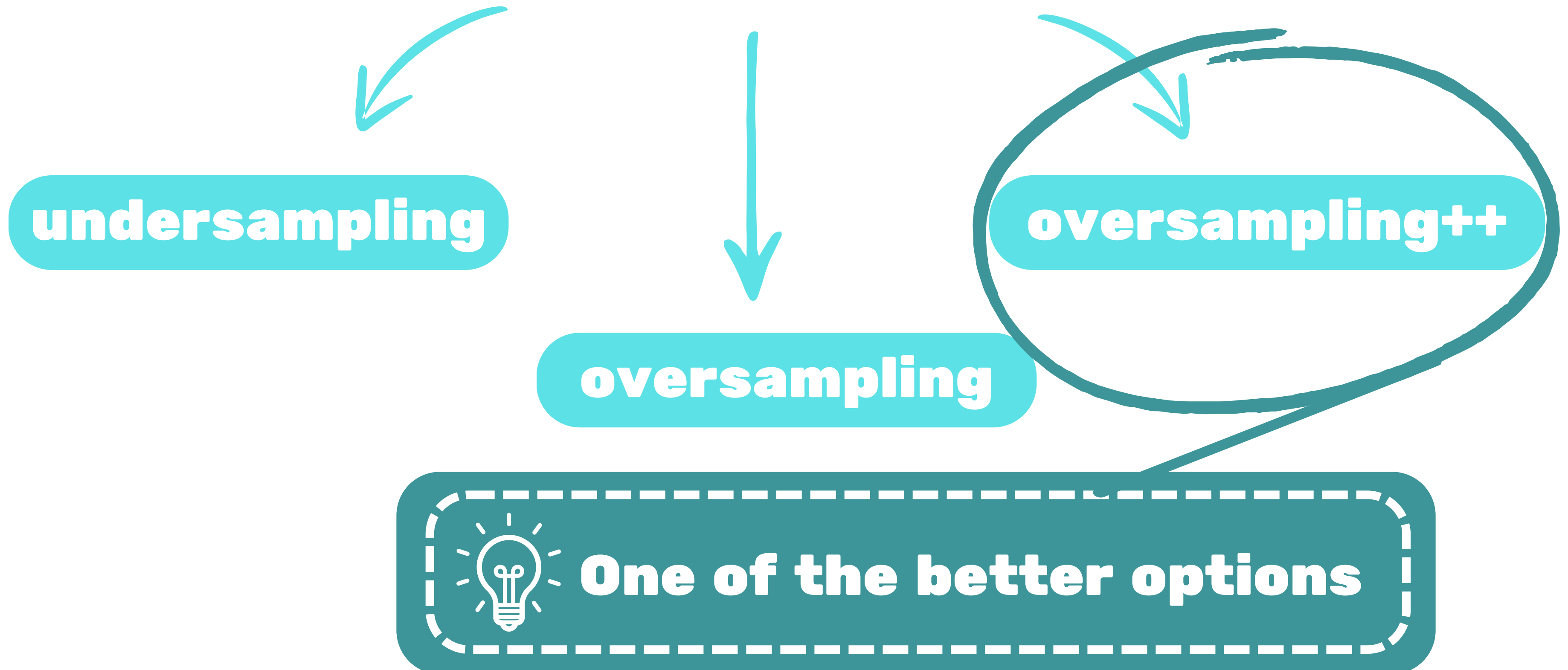


oversampling

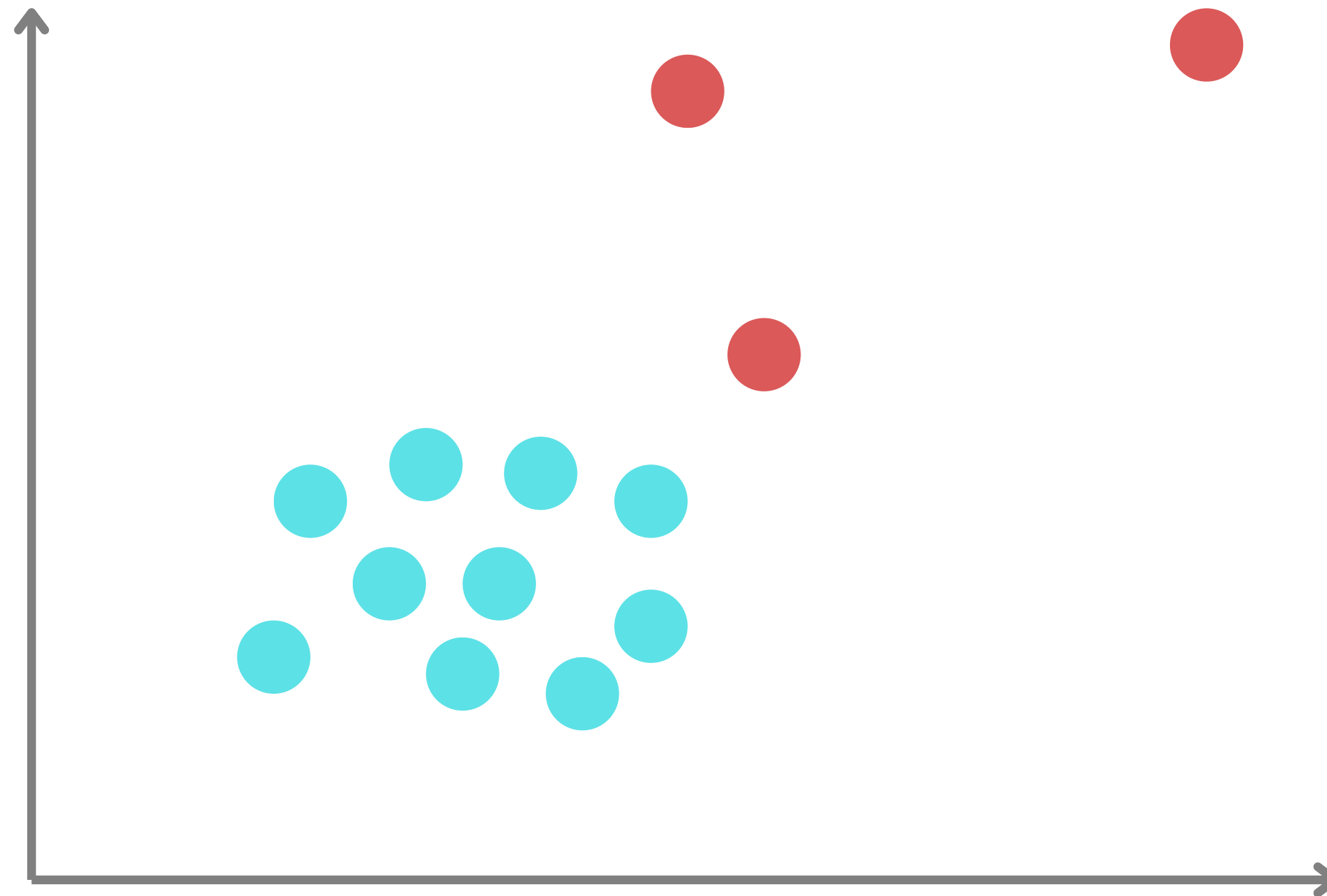


oversampling++

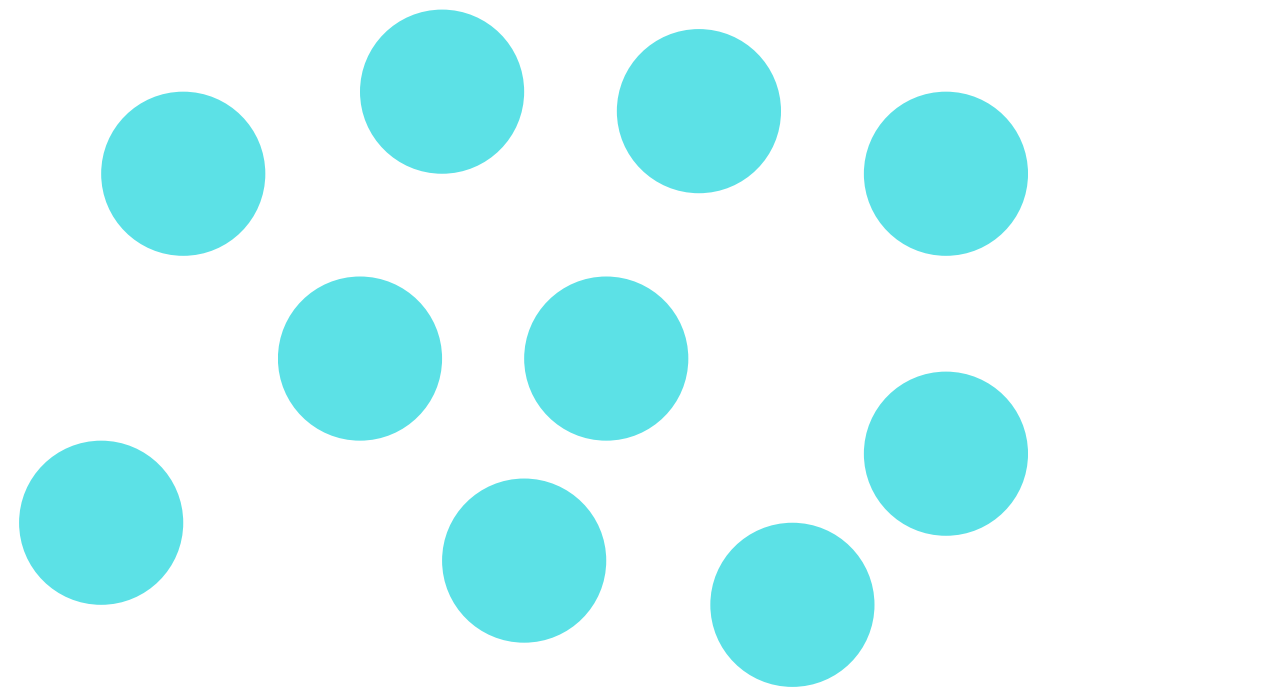
What are our options?

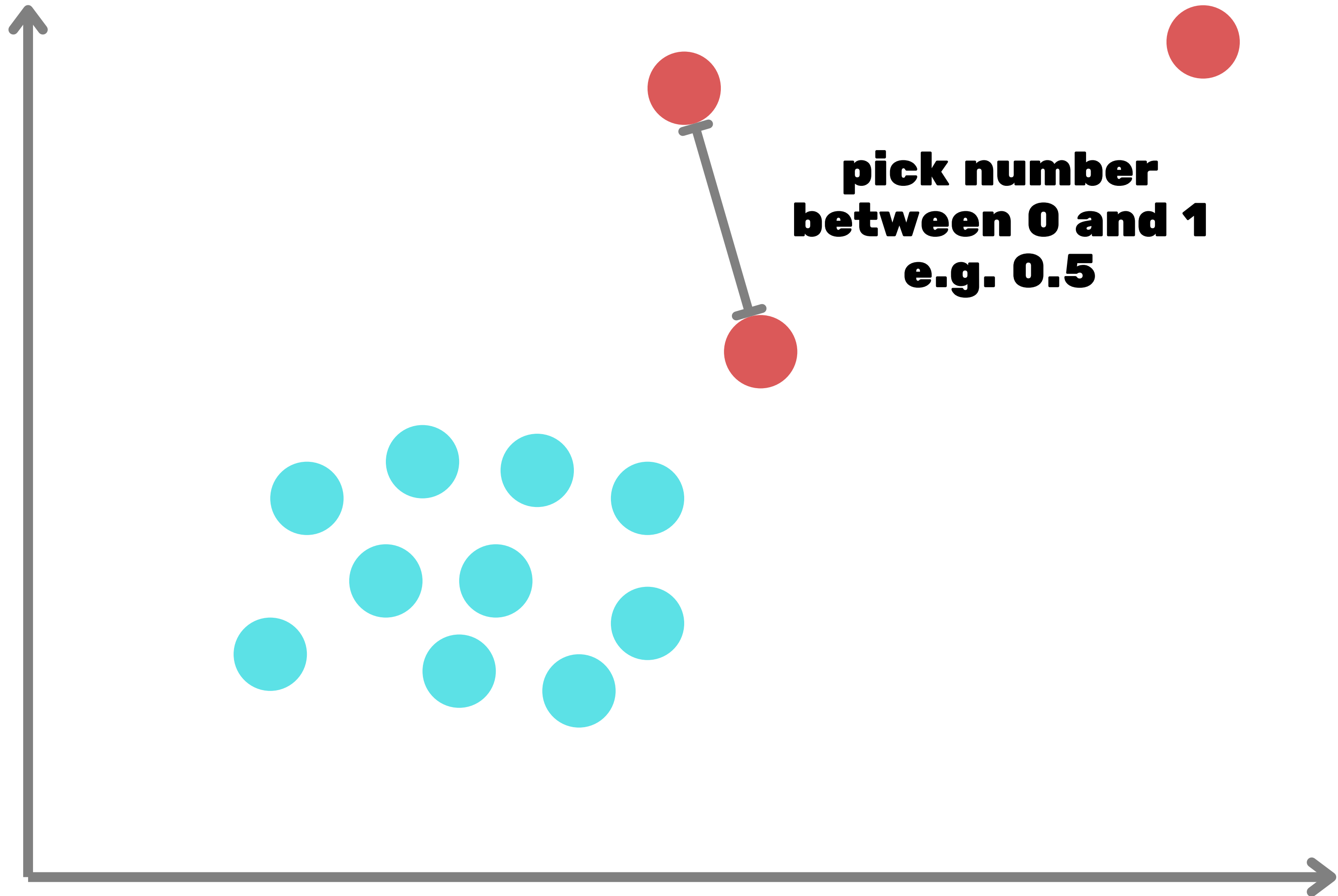


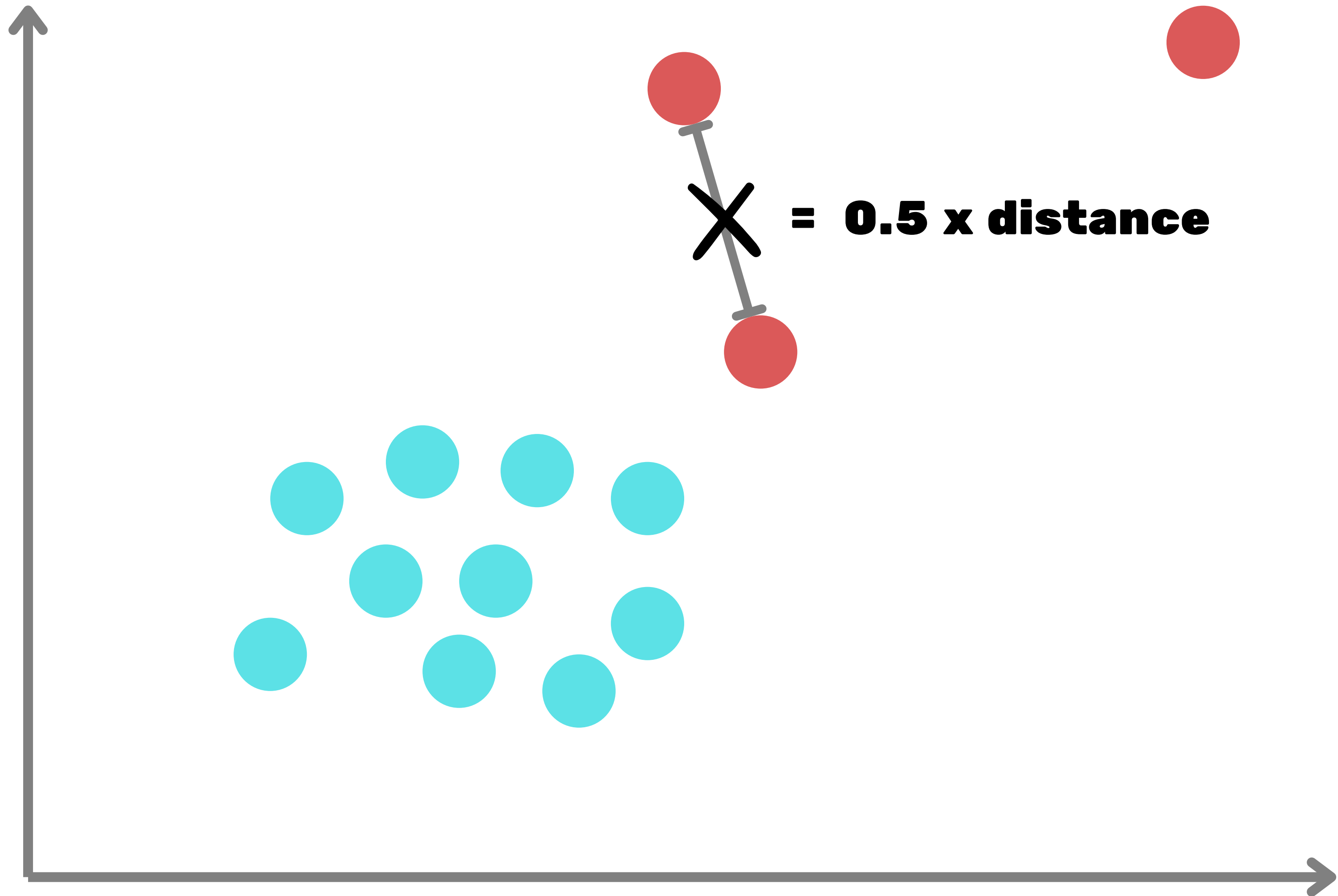
Synthetic Minority Oversampling TEchnique

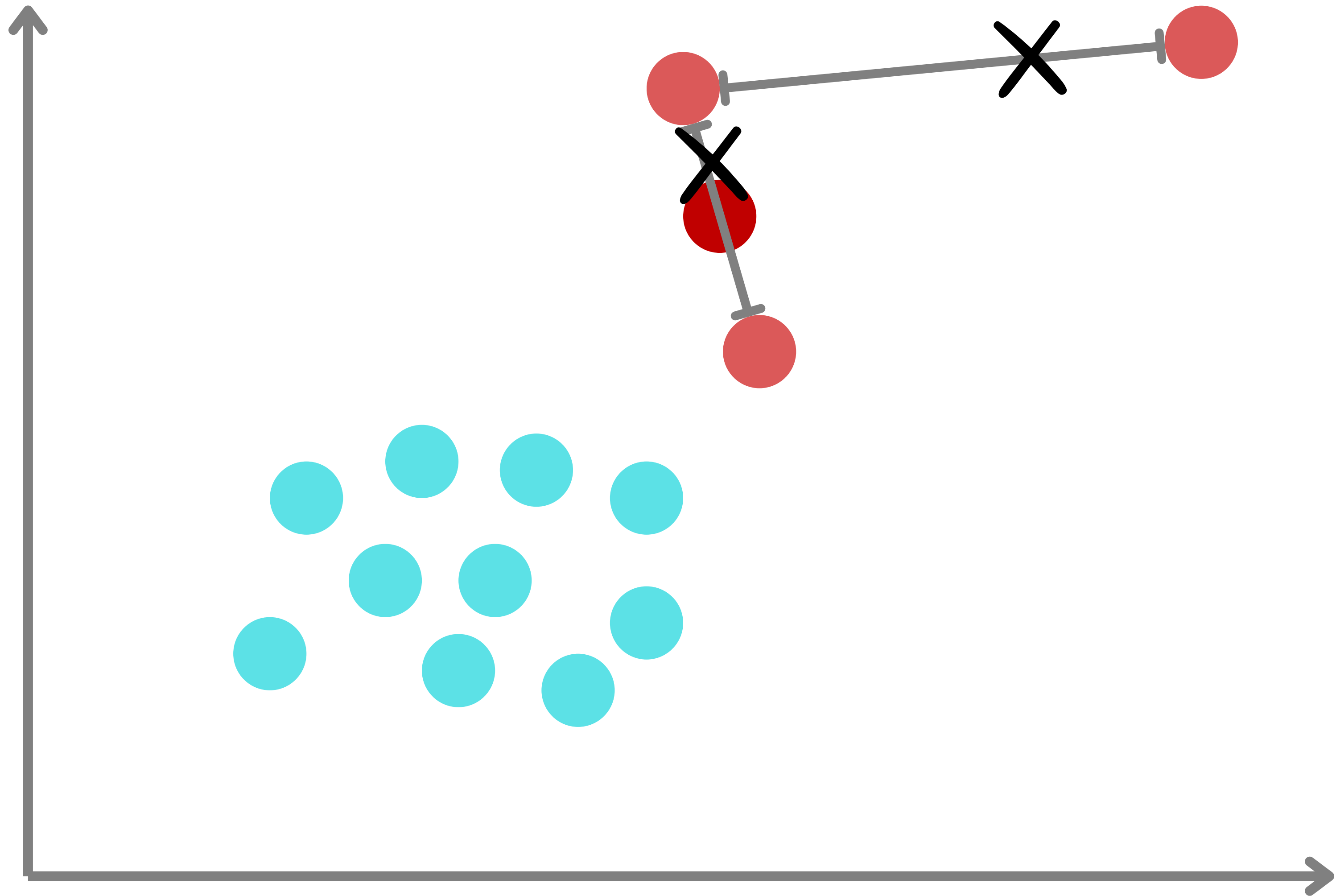


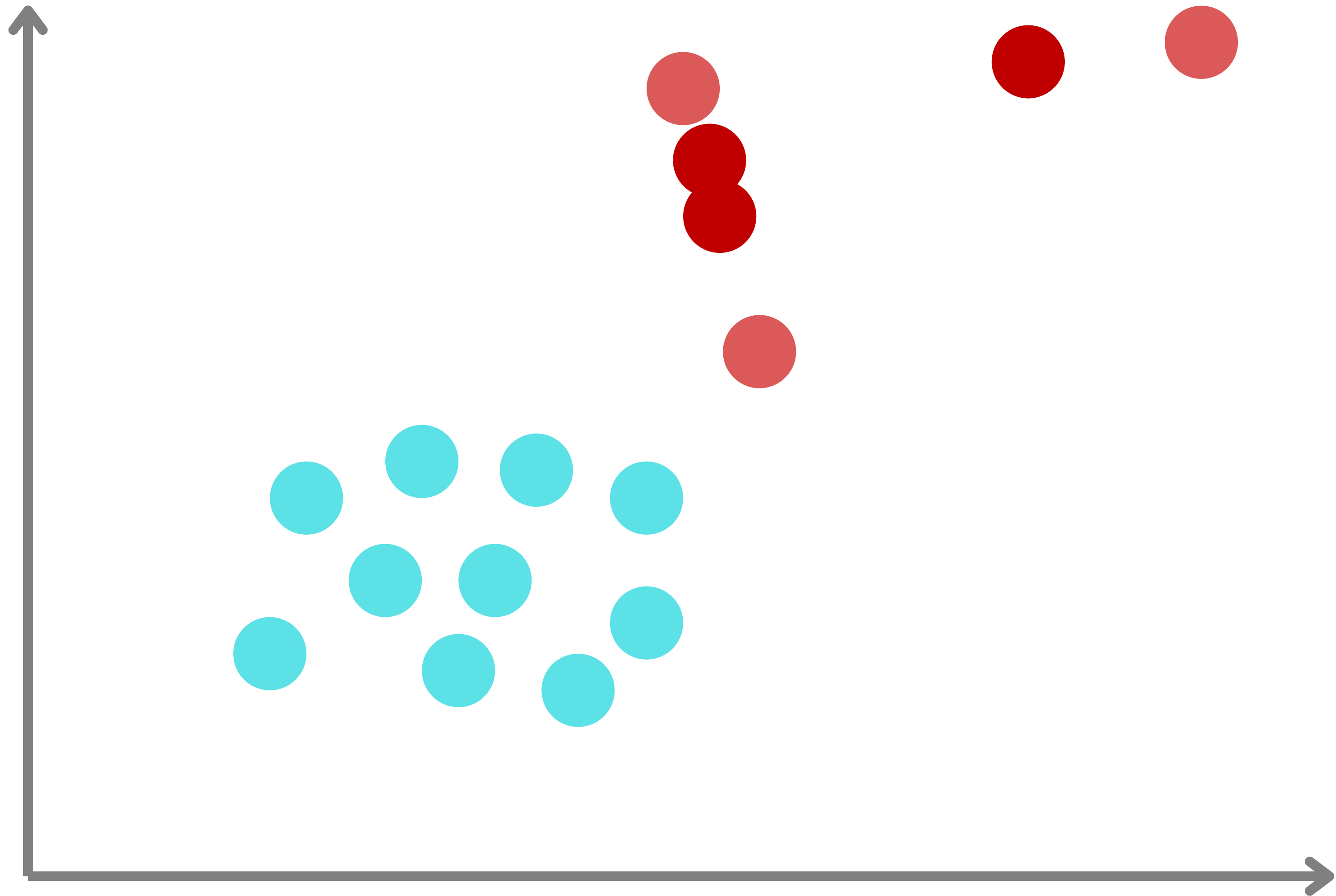
**get distance
from nearest
neighbour**

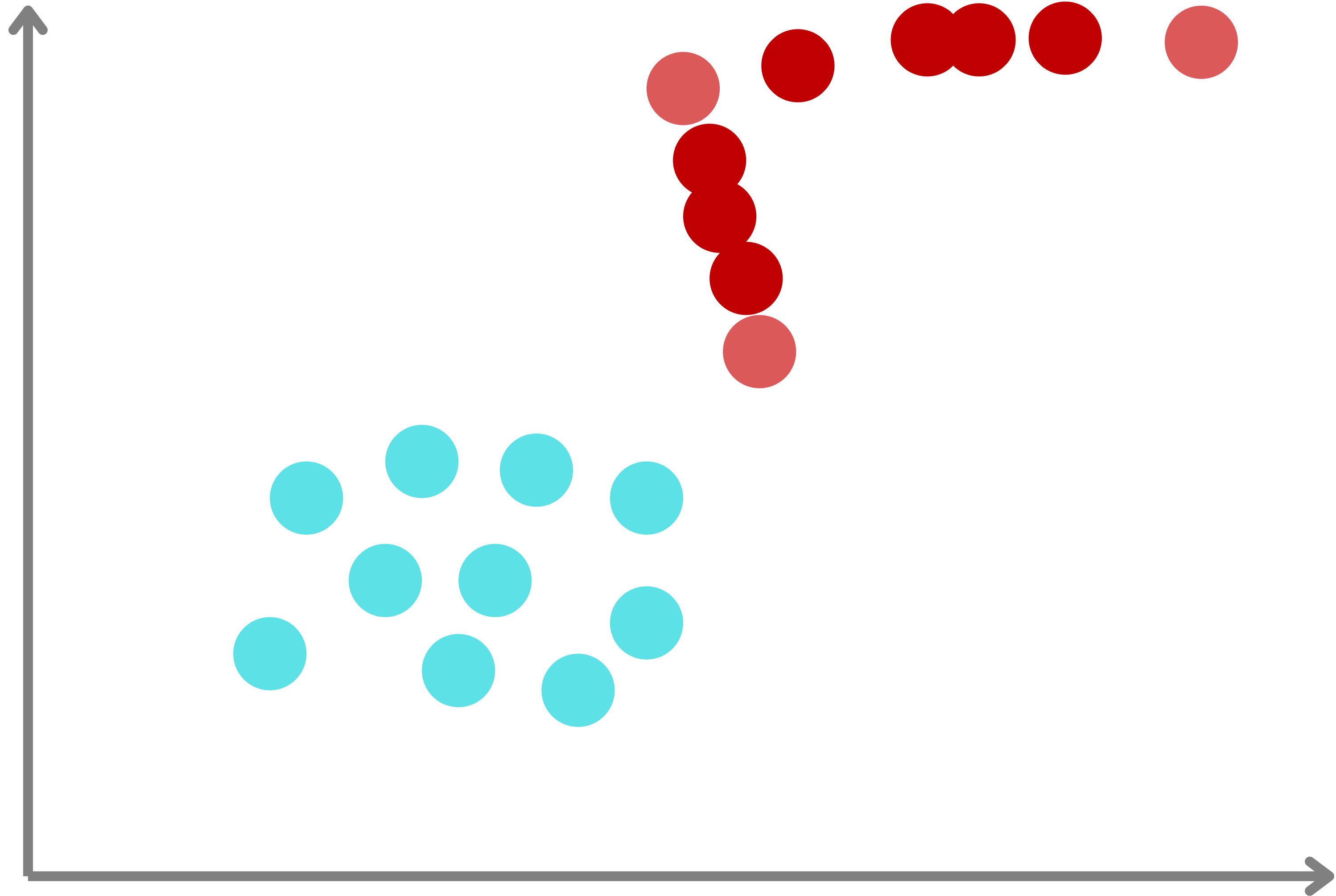


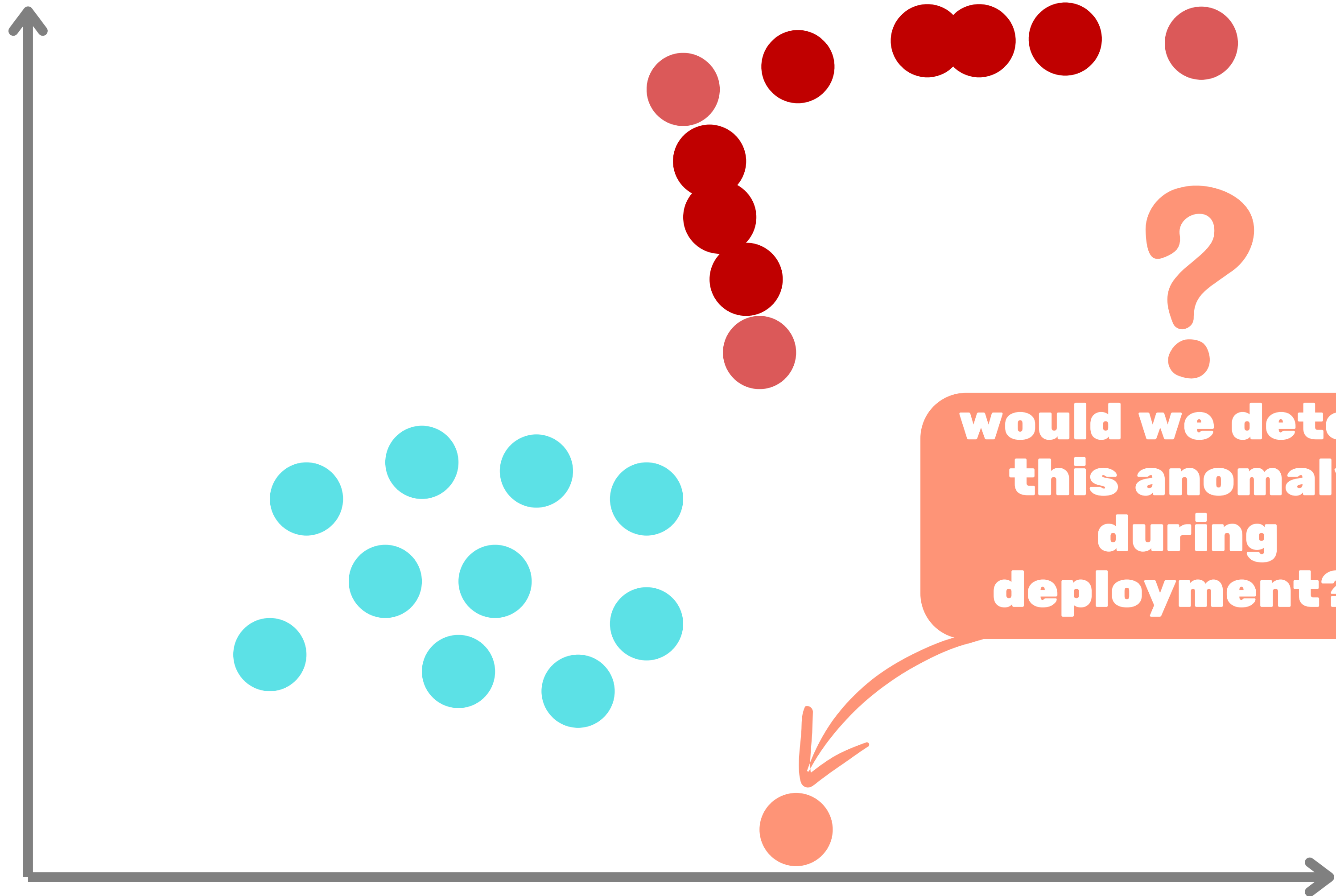












**would we detect
this anomaly
during
deployment?**

Drawbacks



**If imbalance is
really large**



**We won't have a
representative sample**



**If a new type of
anomaly occurs**

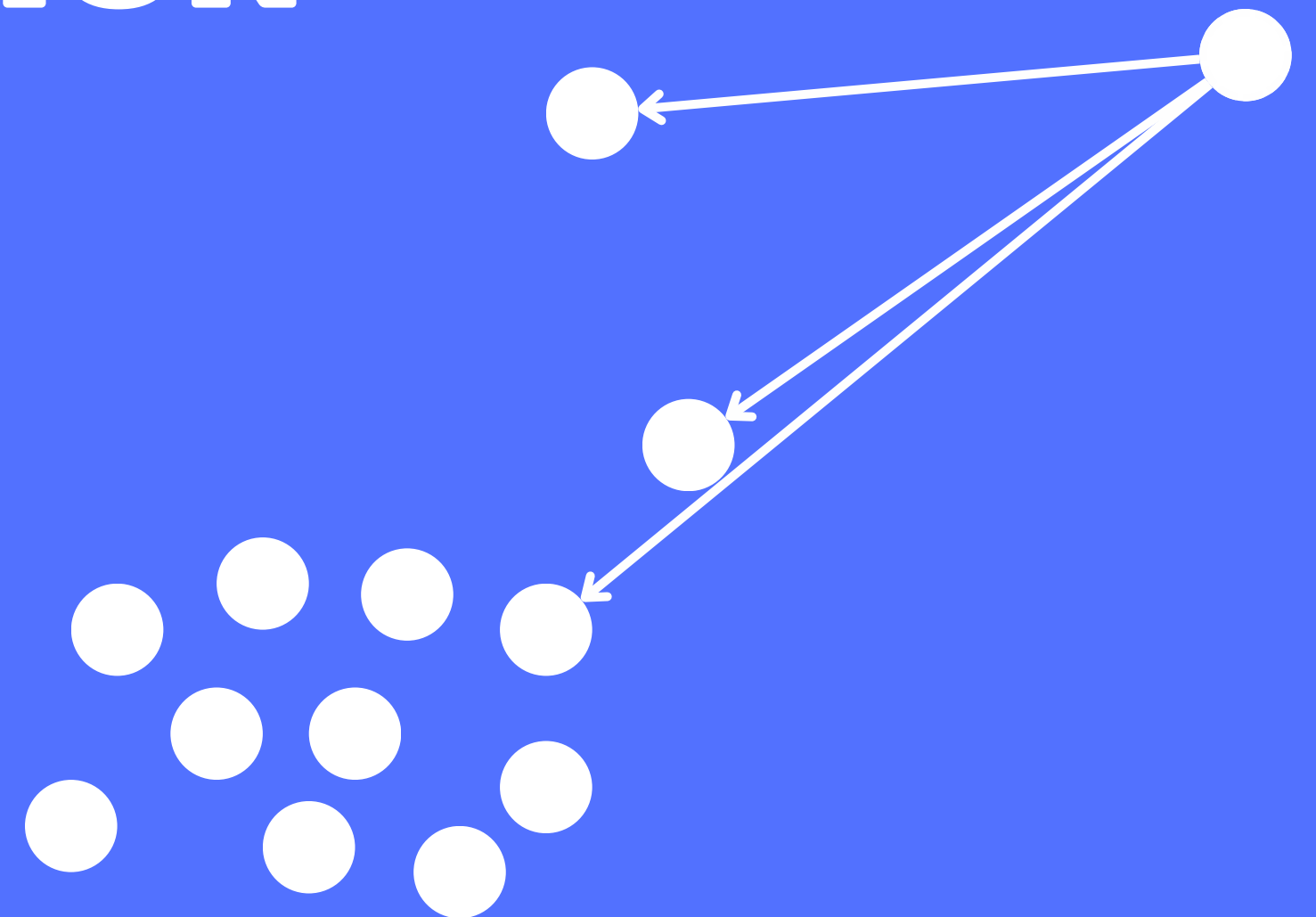
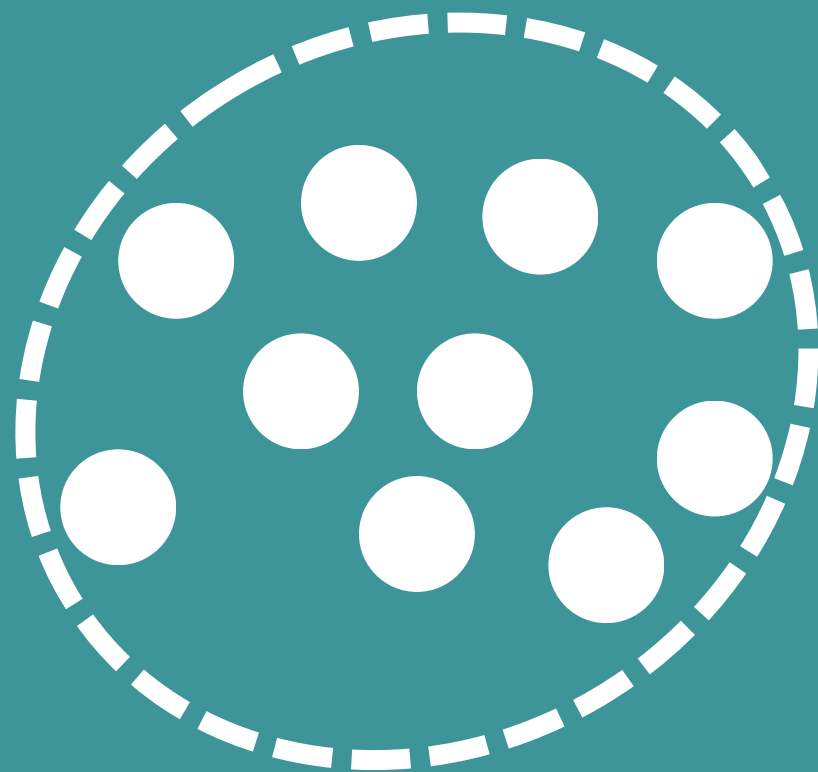


**We might not
detect it**

**SEMI-
SUPERVISED**

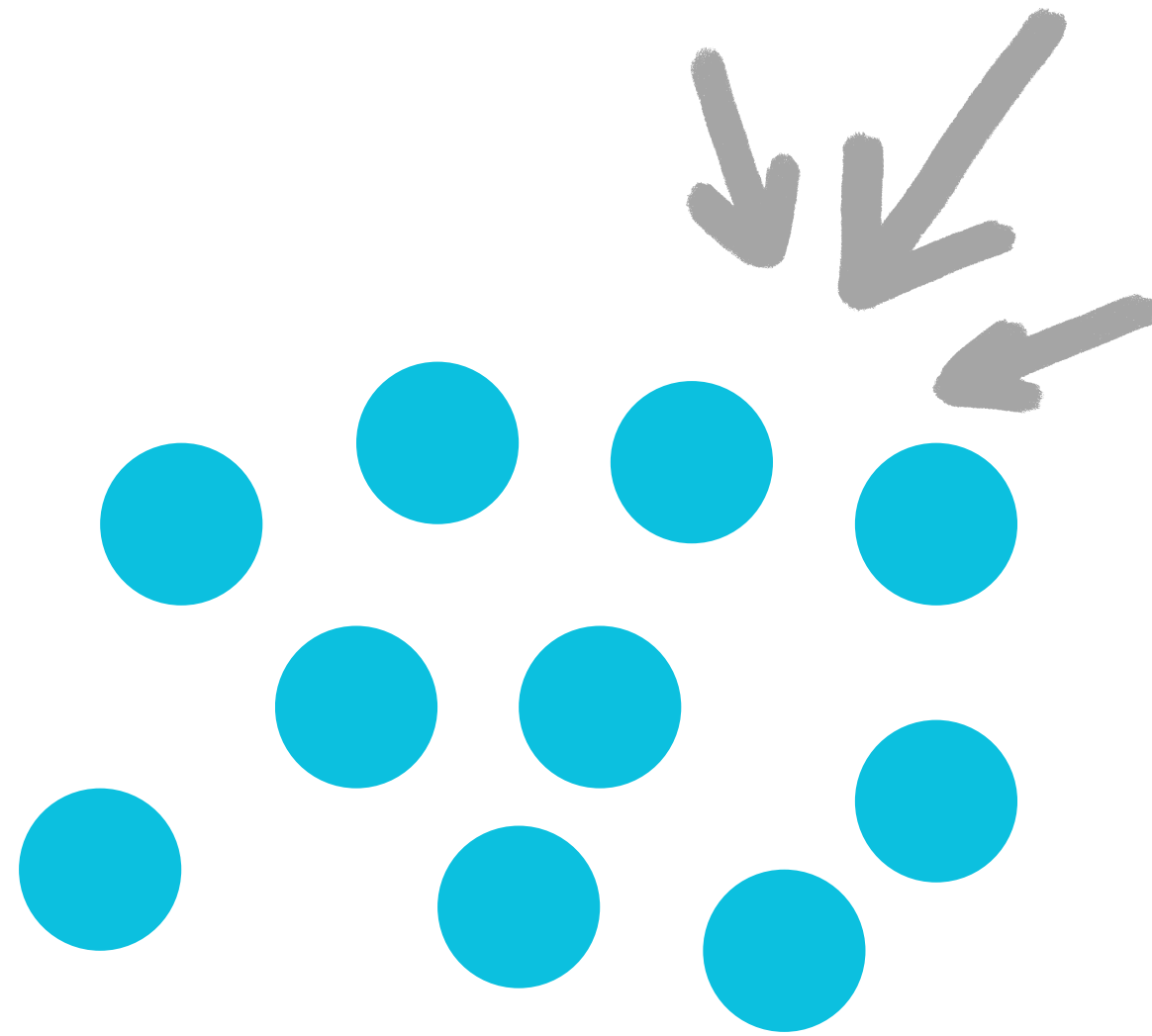
UNSUPERVISED

**ANOMALY
DETECTION**

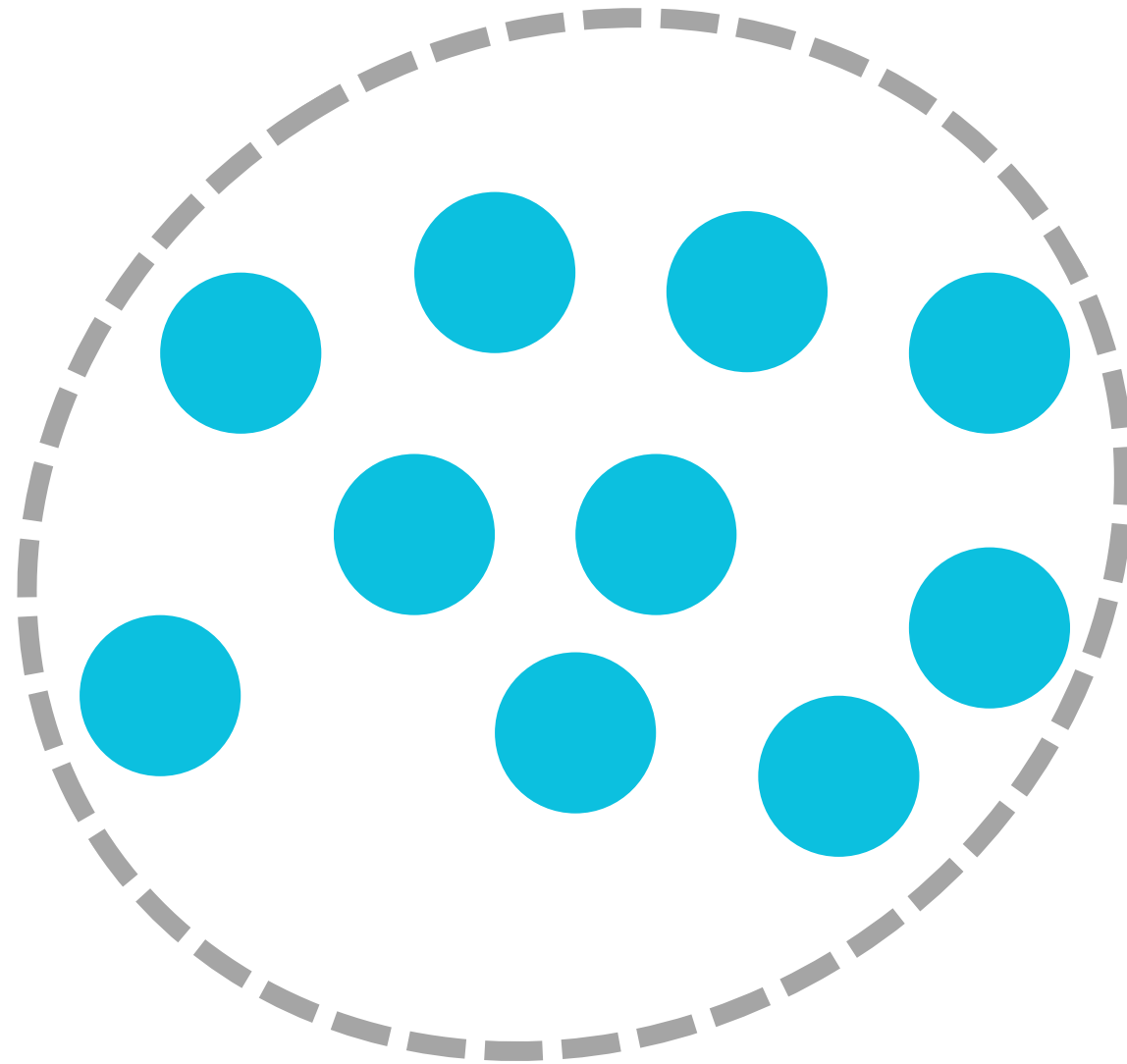


Approach 1
Enclose our normal
data

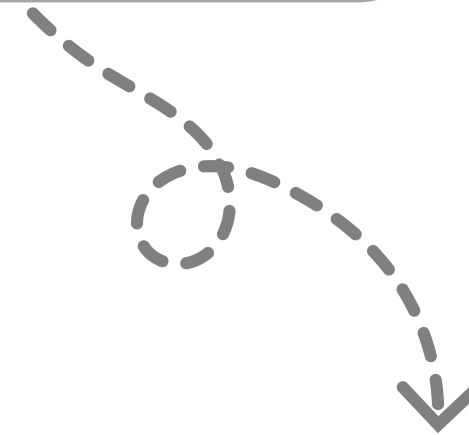
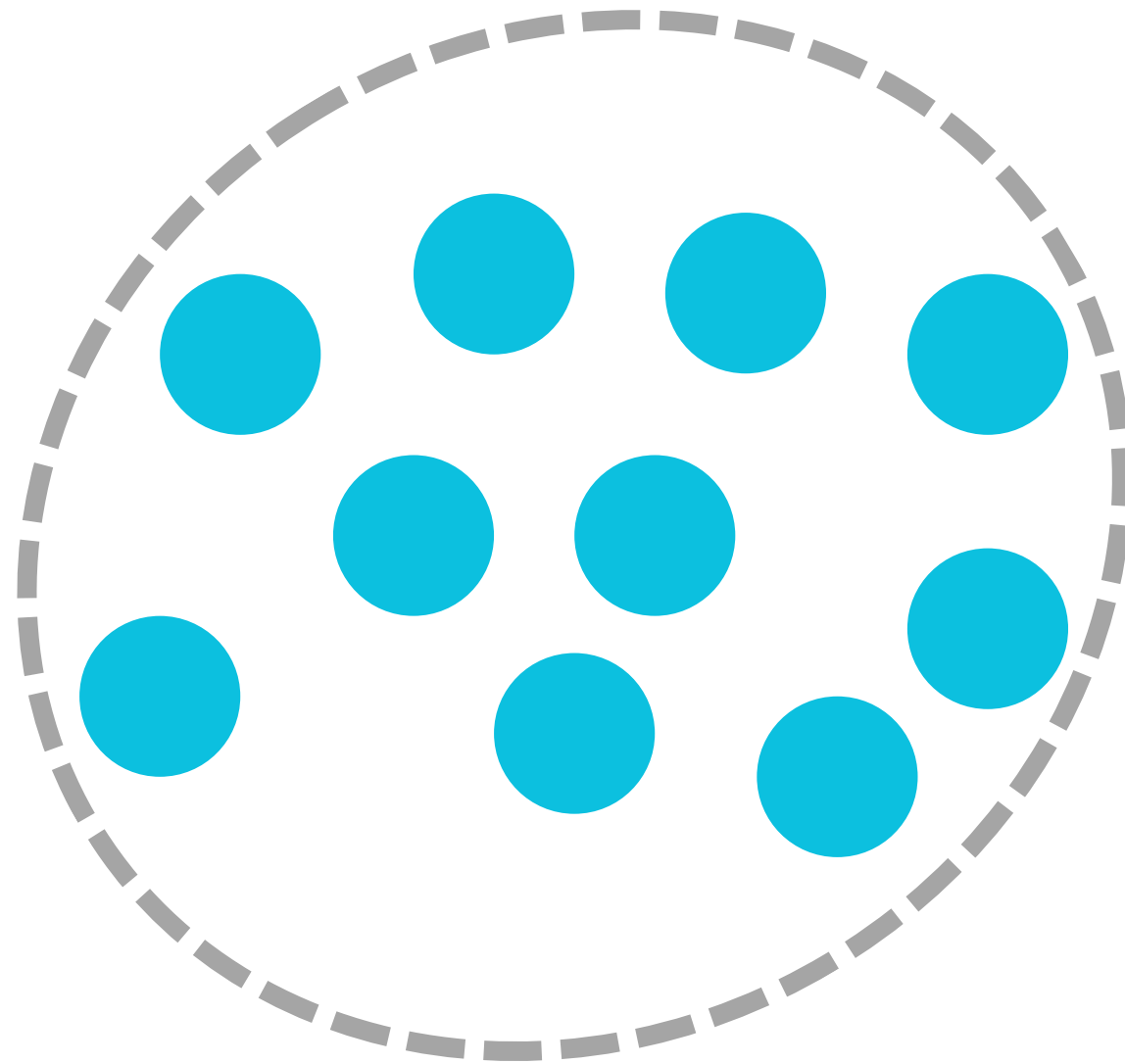
What if we **know** **the data**
we have is **all normal**



we can draw a boundary

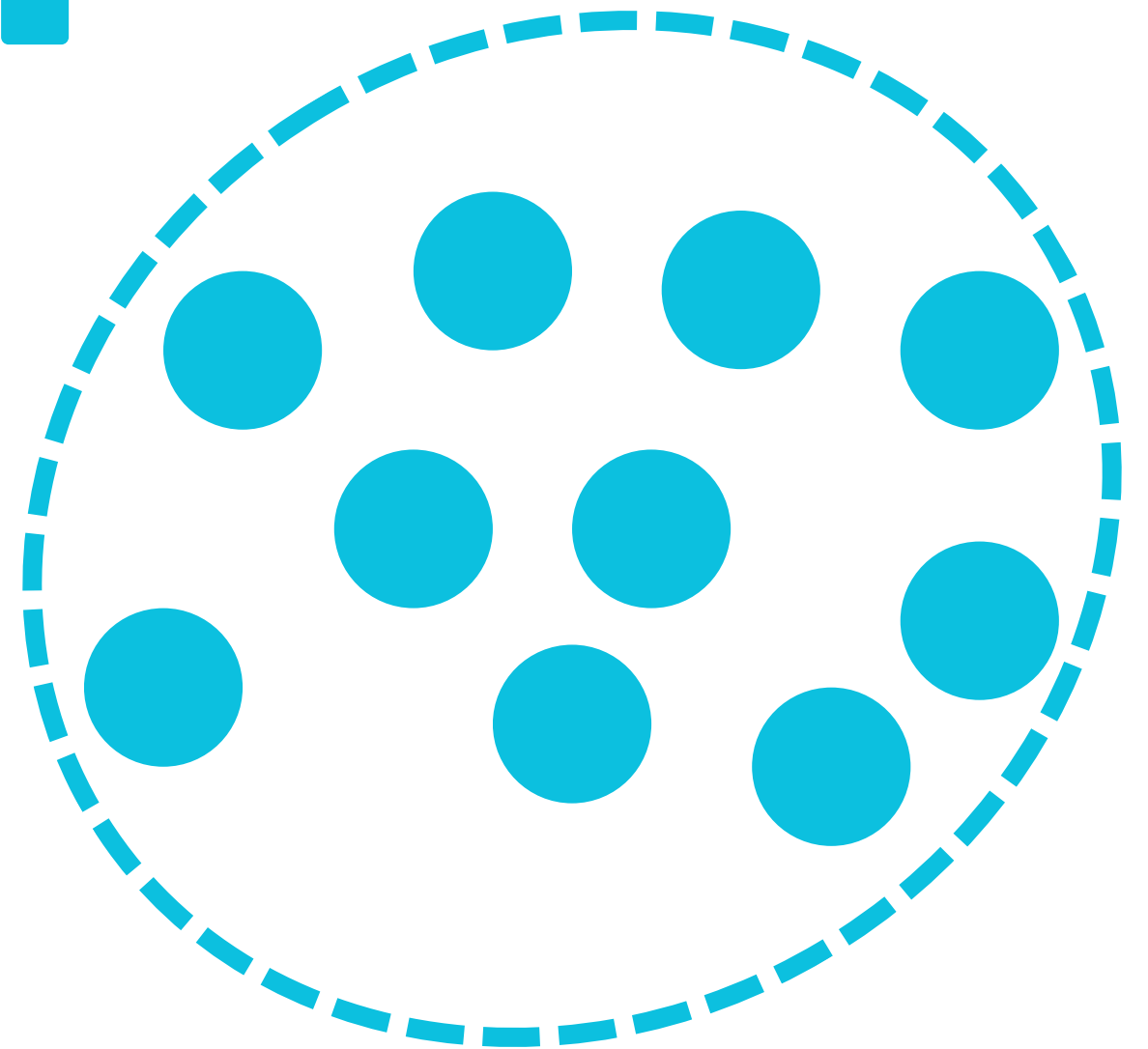


we can draw a boundary around our data



**This is the idea
behind some semi-
supervised methods**

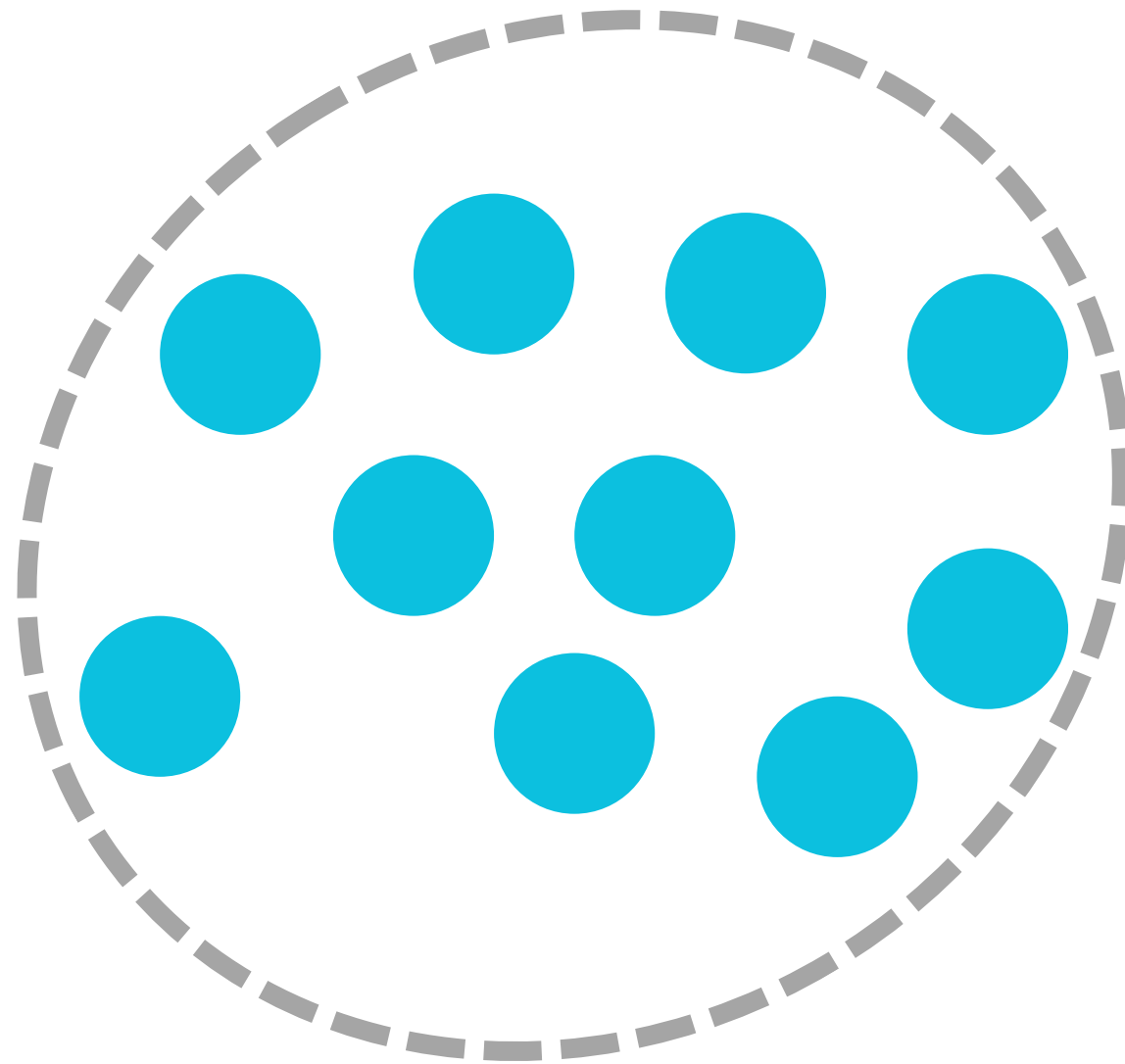
One Class Support Vector Machines (OCSVM)



we aim to

maximise the margin

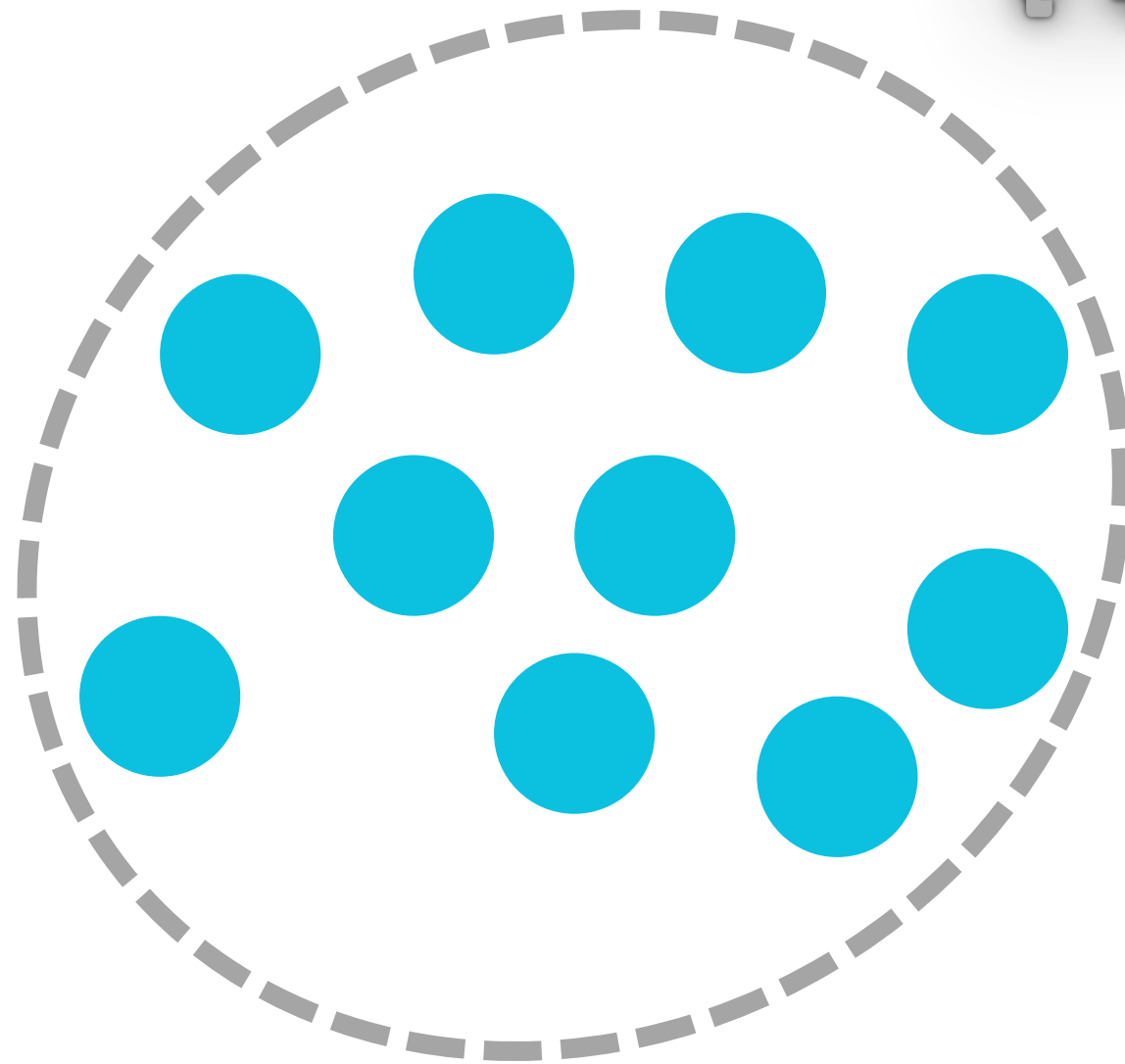
around our normal data



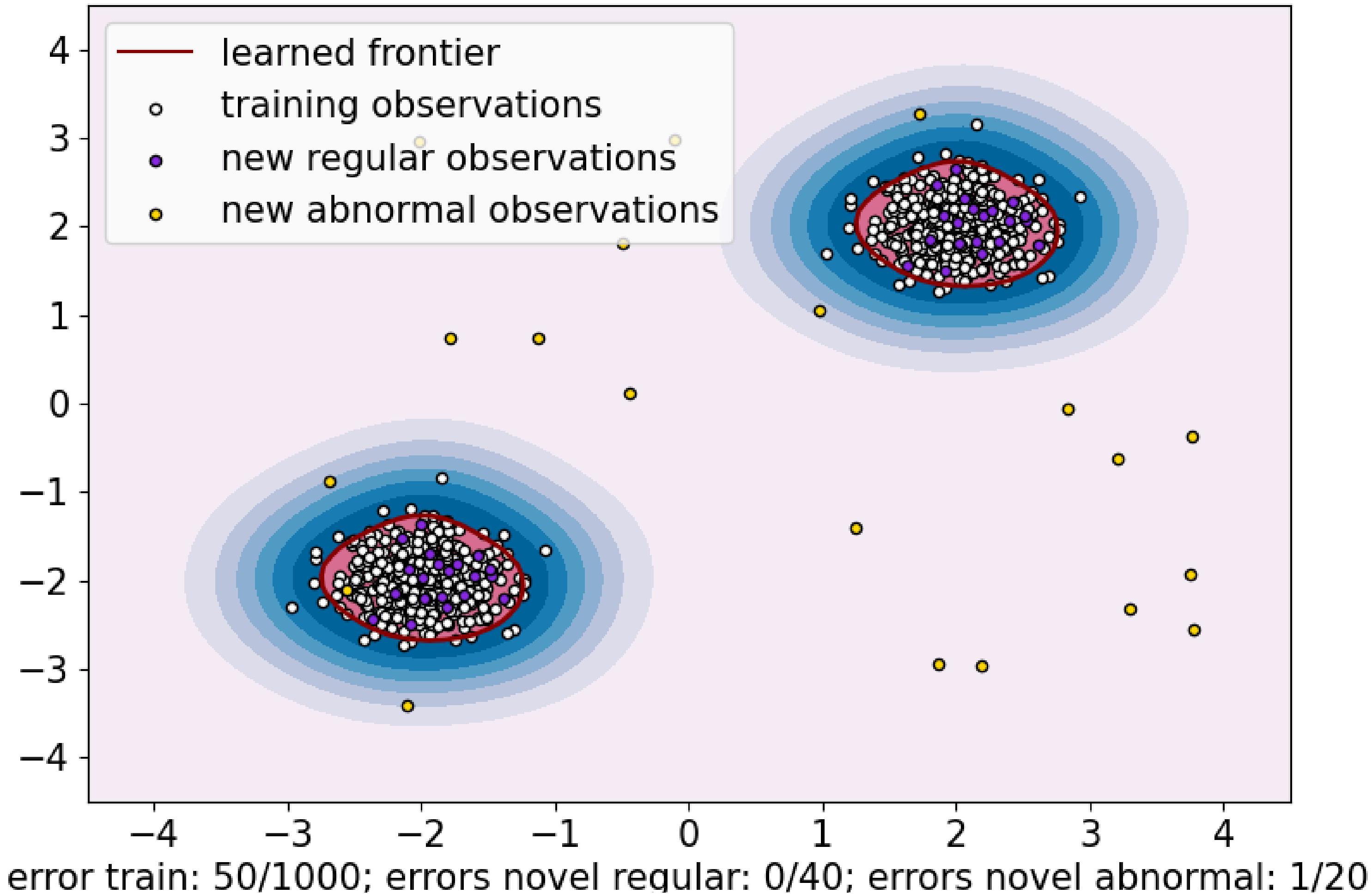
**i.e. find the smallest
circle that encloses our
normal data.**

**In n-dimensional space,
this enclosing "circle"
is called a**

HYPERSPHERE

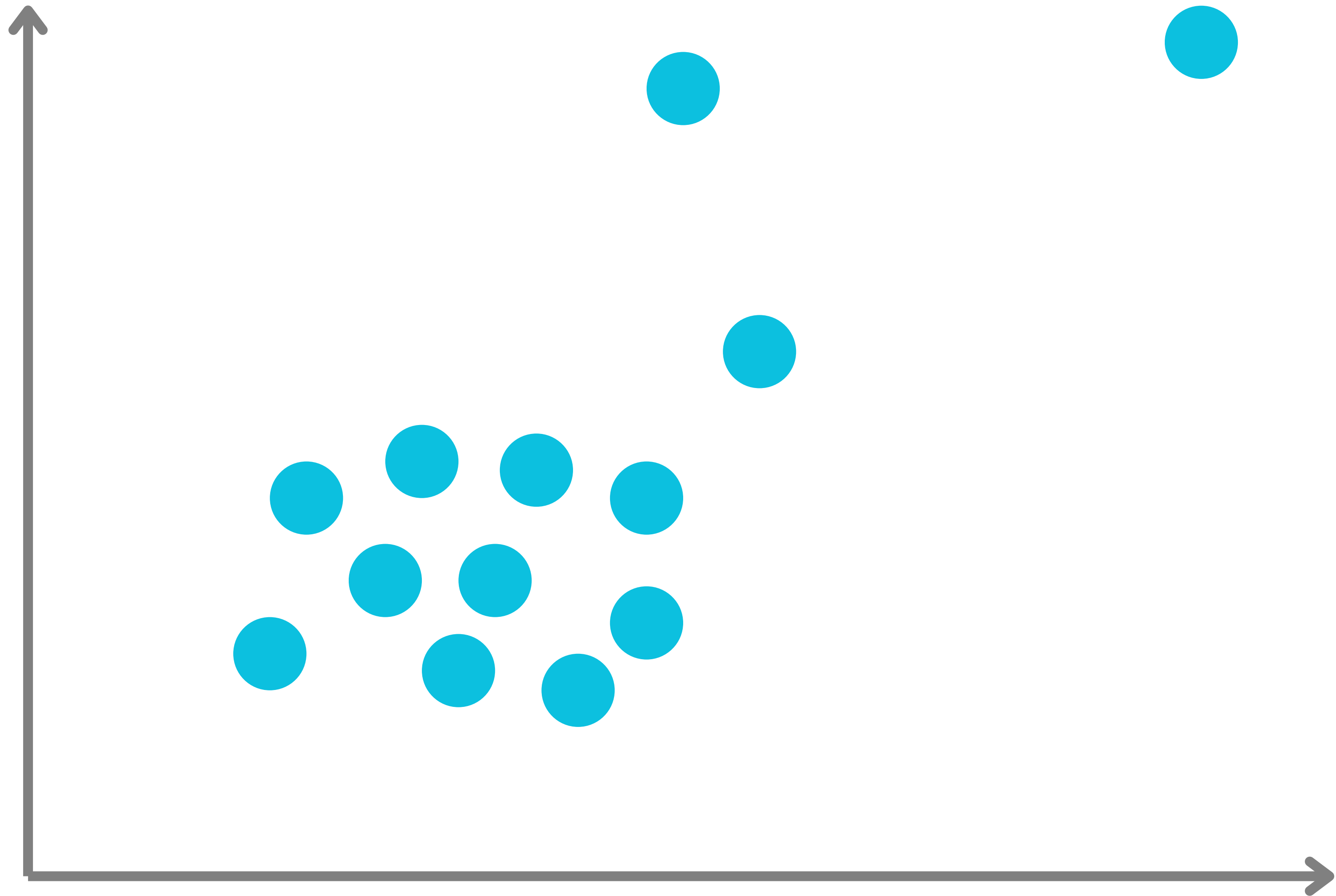


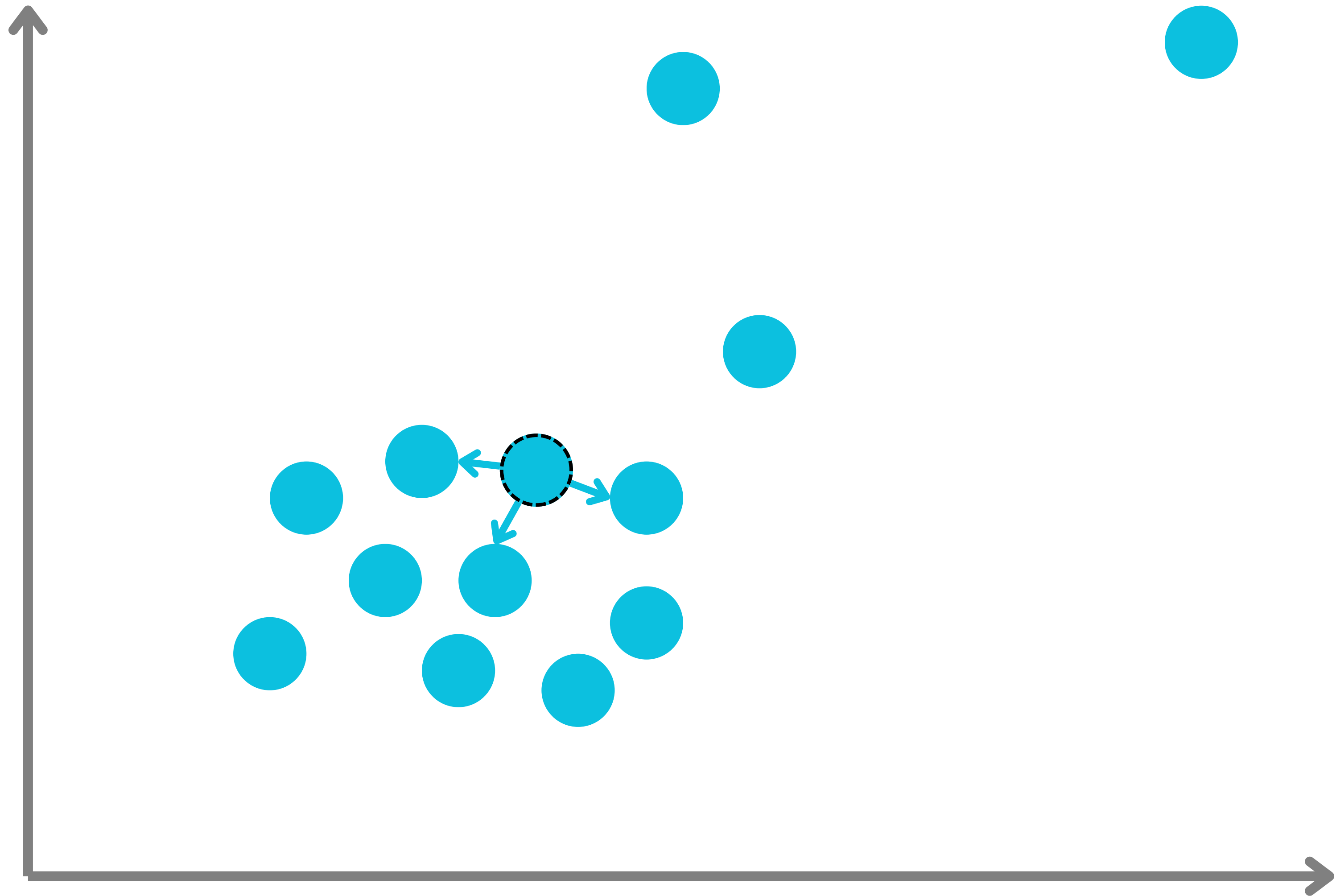
One-Class SVM

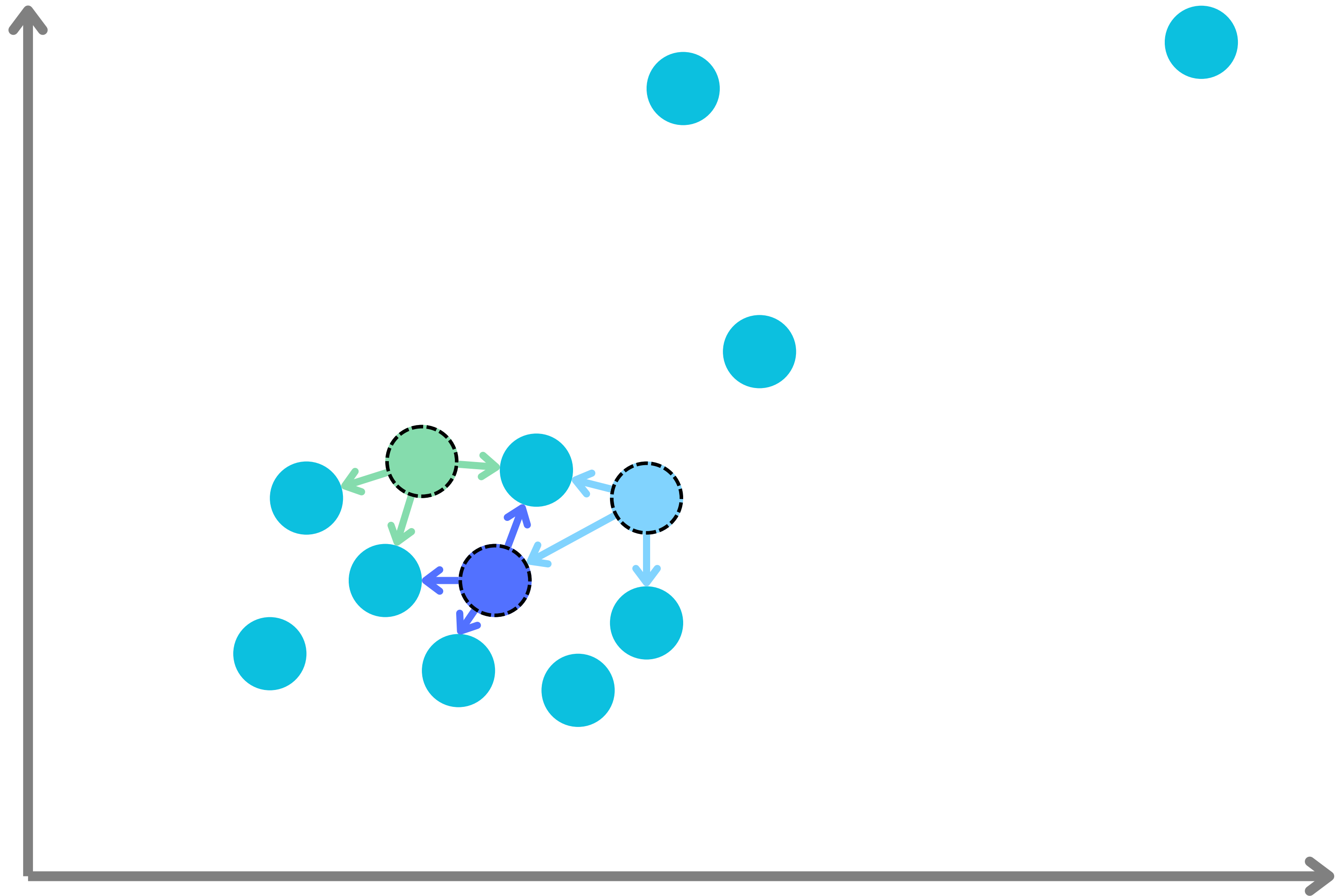


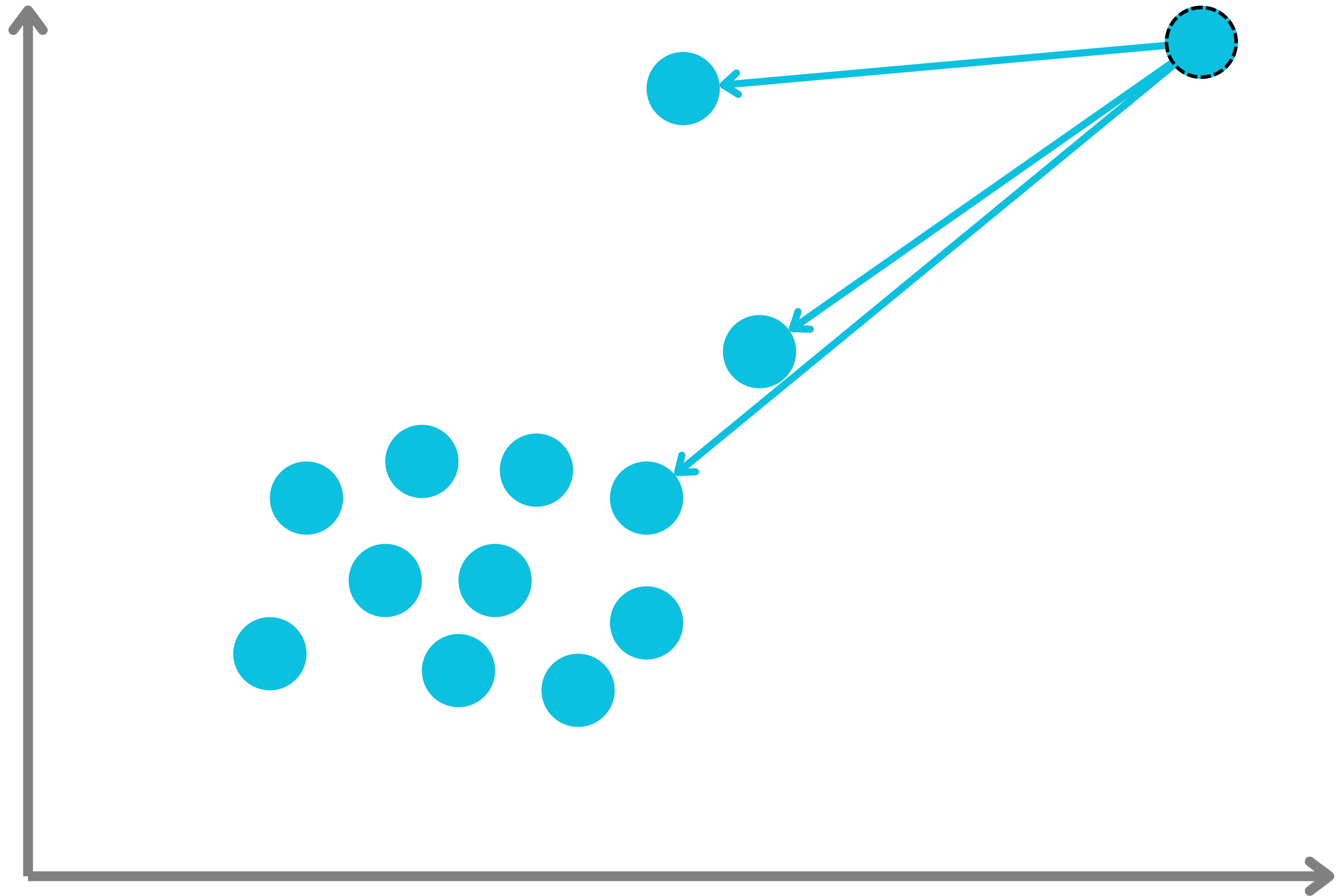
Approach 2

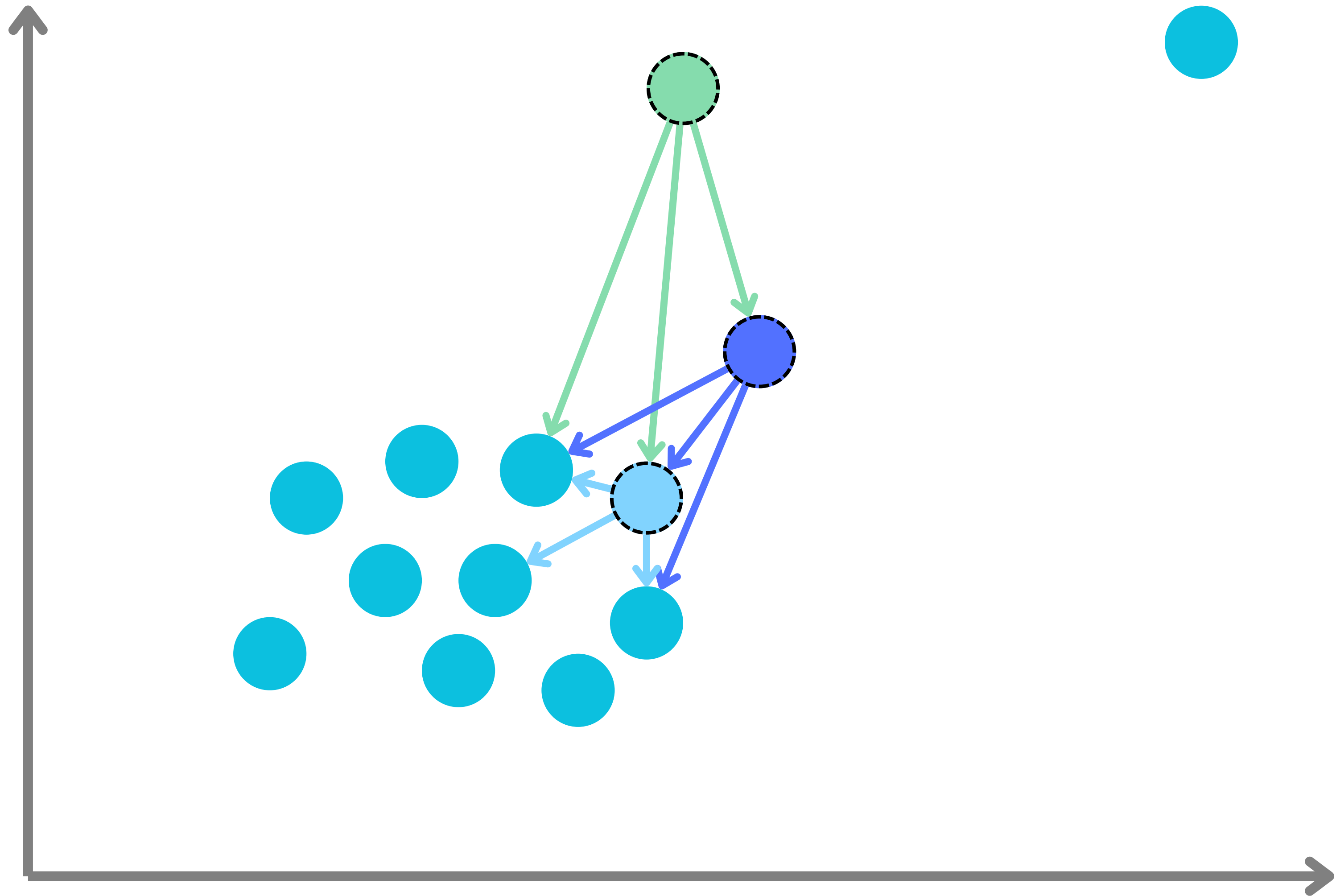
Local density

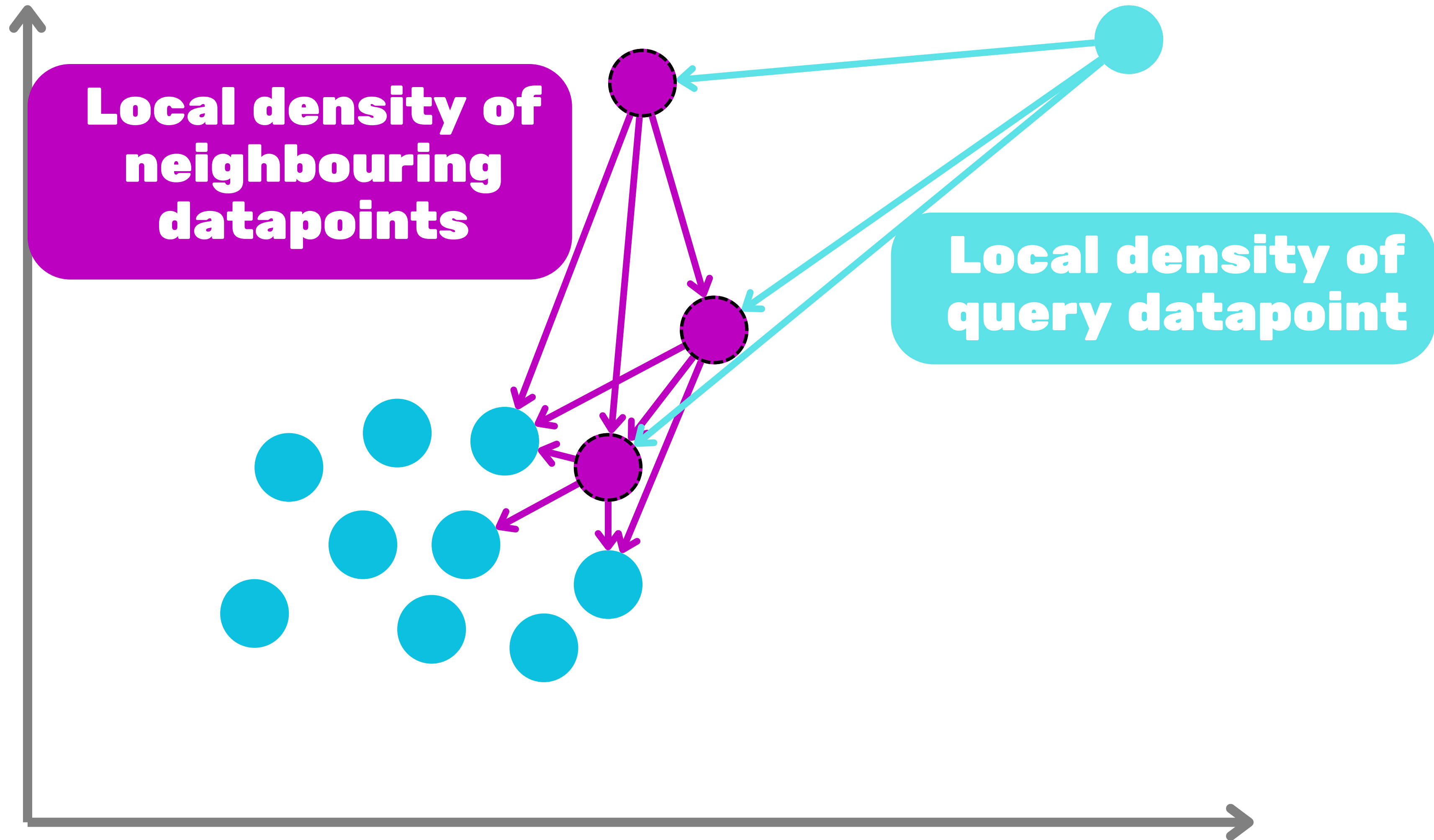












Local Outlier Factor

